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Ion-cut lithium niobate on insulator technology: Recent advances and perspectives

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ABSTRACT

Lithium niobate ($LiNbO_3$ or LN) is a well-known multifunctional crystal that has been widely applied in various areas of photonics, electronics, and optoelectronics. In the past decade, "*ion-cut*" has become the key technique to produce wafer-size, high-quality, sub micrometer-thickness crystalline $LiNbO_3$ thin films, i.e., lithium-niobate-on-insulator (LNOI). With the rapid development of LNOI technology and the tremendous progress of associated surface structuring and engineering techniques over the last few years, many novel chip-integrated $LiNbO_3$-based devices and applications with reduced cost, complexity, power, and size, are demonstrated, boosting the resurgence of integrated photonics based on this material. The remarkable achievements are largely facilitated by the most recent technological progress in photonic integration and performance optimization of LNOI on-chip devices, such as high-quality surface domain engineering, advanced heterogeneous integration technology, powerful dispersion engineering, fine polishing lithography, and wafer-scale fabrication. Accordingly, batch-compatible chip-integrated platforms for more complex photonic integrated circuits, such as quantum optical circuits, are within reach. This article provides a timely review of the key advances in LNOI technology and a reasonable perspective on the near-future directions for both integrated photonics and applied physics communities.

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I. INTRODUCTION

Lithium niobate ($LiNbO_3$ or LN) crystal, being an established choice for existing electro-optic (EO) and nonlinear optical applications over its wideband optical transparency (wavelength range from 400 nm to 5 μm, with an OH-absorption peak at 2.87 μm), also possesses excellent acousto-optic (AO), piezoelectric, and pyroelectric properties, rendering it a unique platform for various applications.[1–4] It is well accepted that the significance of LN to photonics is comparable to that of silicon (Si) to electronics; LN can be therefore acclaimed as "the Silicon of Photonics."[4] In photonics, in addition to the three-dimensional (3D) bulk geometry, optical waveguide structures with diverse geometries based on LN have been successfully applied in integrated systems as well as high-speed modulators, broadband wavelength converters, waveguide amplifiers in telecommunications, and quantum light sources (single photon or entangled photon pairs).[4–12] Despite these numerous achievements, the refractive index contrast (Δn) between waveguide cores and surrounding materials are limited to <0.1 for conventional LN waveguides fabricated by, for example, material-modification based technology such as proton exchange, ion irradiation, Ti ion thermal indiffusion, and femtosecond (fs) laser writing, etc.[5–8,12–14] Large Δn is critical and highly desired with a view to constructing compact devices with high integration level because it enables strong spatial confinement of light field and low bending loss in photonic integrated circuits (PICs).[15–17] Freestanding LN thin films, first fabricated by ion-cut (also named "smart-cut" or crystal ion slicing), possess very high Δn of about 1.2. However, such a thin film possesses a typical thickness of $>10\,\mu$m, which is not suitable for applications in photonic chips due to the highly multimodal light propagation.[18–20] To overcome this bottleneck, a very promising LN thin film configuration called lithium-niobate-on-insulator (LNOI) has been developed successfully on the basis of in-depth and systematic research on a number of key technological processes.[16,18,21–25] The as-prepared LNOI thin film usually consists of an ultrathin LN layer with well-preserved bulk properties, a sufficient low-index cladding layer (usually an insulator layer) bonded to the LN layer to provide a large Δn, and a supporting substrate holding the bonded LN/cladding layer.[18,26] Due to its large-size, high-quality, and large refractive index contrast, LNOI thin film has triggered evolutionary advancement of LN photonics, enabling simple scaling of photonic devices/circuits down to the sub micrometer scale while simultaneously well-preserving the excellent optical properties of bulk LN. In fact, ion-cut technology was initially developed for silicon-on-insulators (SOI) technology and later imitated to ferroelectric crystals,[27–29] representing a very universal technology that enables transferring very fine layers of crystalline materials from a donor substrate to an acceptor substrate (can be a totally different material), overcoming physical limitations, and opening up new opportunities and possibilities for thin-film technology. In principle, ion-cut technology enables any thin-film material (namely "X" material, e.g., LN, $LiTaO_3$, 4H-SiC, GaN, Ga_2O_3, etc.) to be transferred on top of any other materials (usually insulators) while maintaining the initial crystallographic, and thus optical, properties, which is very promising for construction of hybrid "XOI" photonic platforms for future multifunctional PICs operating in both classical and quantum regimes.[30–40] It is worth mentioning that thin-film LN with thicknesses of hundreds of nanometers to several micrometers can be achieved either by an ion-cut or by wafer thinning technique (such as ion milling or direct wafer lapping/polishing); both technologies are able to produce high-quality LNOI thin films.[16,18,23,25,26,41–44] In contrast to the latter case, nevertheless, ion-cut technology avoids waste of the raw LN materials, which can be reused for many times. The discussion in this review article will focus on ion-cut LNOI technology.

Over the past few years, with the rapid development of LNOI technology and the tremendous progress of associated surface structuring and engineering techniques, many novel chip-integrated LN-based devices and applications with reduced cost, complexity, power, and size, are demonstrated, boosting the resurgence of integrated photonics based on this material.[16,18,21–25,45–73] Fairly speaking, the LNOI technology and many novel on-chip micro/nanoscale photonic devices based on it are revolutionizing both the LN industry and the modern integrated photonics, enabling entirely new devices with high performance and rapidly emerging applications with great potential. It is also for this very reason, in 2017, a bulletin titled "*Now Entering, Lithium Niobate Valley*" was published, highlighting the significant breakthroughs achieved on LNOI for integrated photonics and in parallel foreseeing the birthplace of photonics revolution.[74] In addition to photonics, the thin-film geometry of LN brings out cost-effective electronic devices as well, such as infrared detectors and ferroelectric memories.[75–77] Ever since the market introduction of high-quality, large-size thin-film LN commercial products, this field has significantly proliferated, as implied by the increase in the quantity and diversity of publications, especially over the last few years.

In addition to LNOI products, particularly, the remarkable achievements of LNOI photonics are also facilitated by the most-recent technological progress in photonic integration and performance optimization of on-chip devices, such as high-quality surface domain engineering, advanced heterogeneous integration technology, powerful dispersion engineering, fine polishing lithography, and wafer-scale fabrication.[78] Accordingly, batch-compatible chip-integrated platforms for more complex PICs, such as quantum optical circuits, are within reach.[79,80]

There have been a few review articles on LNOI-based photonic devices, summarizing the research achievements obtained in many aspects.[18,21–25,81] This article aims to provide a timely review of the key advances in LNOI technology and a reasonable perspective on the near-future directions. Following Sec. I, a brief summary of the development of LNOI technology is given, concentrating on three key techniques, namely ion implantation, defect engineering, and wafer bonding, that are involved in LNOI fabrication. Then, the up-to-date fabrication and performance of LNOI on-chip optical components, including waveguides, micro-cavities, and photonic crystal (PhC) structures by a series of advanced micro-/nano-structuring methods are overviewed. The following section is divided into four parts according to the most-concerned technological advances: surface domain engineering (concentrating on fabrication of quasi-phase matched structures, namely periodically poled LNOI or PPLNOI thin films), integration of active ions (mainly rare-earth ions), implementation of heterogeneous systems (focusing on the hybrid systems containing LN and CMOS-compatible materials, plasmonic structures, quantum dots, etc.), and optical coupling strategy (introducing the widely used methods for efficient optical coupling between the standard optical fibers and LNOI on-chip photonic devices). In each section, correspondingly, the representative demonstrations of LNOI-based photonic devices are summarized, highlighting the emerging applications and concepts without losing emphasis on the latest breakthroughs. Moreover, a considerable length of each section will discuss how each technology may shape the future directions of LNOI photonics and even the whole integrated photonics field, providing useful perspectives on the short- and long-term future of this very active topic. This review article will conclude with a short summary as well as perspectives on some well-noted topics and rising applications of LNOI photonics that are not involved in the previous sections.

II. DEVELOPMENT OF ION-CUT LNOI TECHNOLOGY

The well-developed ion-cut LNOI technology is based on three core techniques: ion implantation, wafer bonding, and chemical-mechanical polishing (CMP).[18] During the whole preparation process, it is highly desired that the LN thin film remains single-crystalline with a view to preserving original optical features of bulk LN. The removal of defects and damages induced by ion implantation is therefore very necessary for practical applications of LNOI in photonics.[3,26] For integration of more optical components on a single chip, a large-scale LNOI wafer with a uniform thickness is desirable. So far, up to 4-inch-diameter wafers have been successfully produced and become commercially available.[18,23] Some parameters, such as surface roughness, thickness uniformity of wafer-scale LNOI thin film, and flatness of the SiO_2-LN bonding interface, are crucial for high-quality wafer fabrication as well. A detailed fabrication process of LNOI thin films was first proposed by Günter's group at ETH Zurich and has been updated and summarized elsewhere.[18,82] But for completeness and convenience, a brief introduction of the production flow is given here. Figure 1 schematically illustrates the typical 5-step fabrication of LNOI wafer, with LN as the supporting substrate. First, He^+ ion implantation at an energy of 200–400 keV is performed on a LN wafer (donor wafer), creating a buried damage layer at the end of the ion range. Second, a layer of SiO_2 with a thickness of a few micrometers is deposited onto another LN wafer (acceptor wafer). Third, the donor and acceptor wafers are directly bonded together at room temperature. Fourthly, under thermal annealing treatment at moderate temperature (about 150 °C–250 °C), a thin layer of LN is lifted off from the donor wafer and bonded onto the acceptor wafer. Finally, with careful CMP of the LN thin film surface, LNOI wafer with a typical thickness (i.e., thickness of the active LN thin film) of 300–900 nm can be fabricated. Furthermore, a post-fabrication high-temperature thermal annealing treatment (generally at 400 °C–500 °C) of LNOI wafer is usually required to further improve the quality of as-prepared LN film before final fine CMP process. The final thickness of LN thin film is determined by the energy of implanted He^+ ions and the surface polishing. The fundamentals and technological points are introduced in Subsections II A–II C.

A. Ion implantation

Ion implantation (or ion irradiation in the case of heavy ions) is a well-established material-modification technology applicable to a variety of solids.[6,13,26,83–85] It is a "physical" solution to induce a number of new ion-matter interaction and radiation damage effects to artificially tailor the properties of target materials. The charged ions are accelerated by ion implanters or accelerators and bombarded onto the target, losing their energies through a series of interaction with target atoms. The process of ion energy deposition into targets also locally modify material properties through changing the material structure and generating more extended defect structures.[6,26] So far, ion implantation/irradiation has been widely applied in a large variety of solids, e.g., metals/alloys, dielectrics, low-dimensional materials, or biologic materials, toward versatile applications covering a wide range of research fields.[6,26,83] The possible material modification effects that can be induced in solids include, by adjusting ion species and implantation parameters, localized defects creation, selected alien ion doping, nanoscale particles synthetization, as well as surface morphology patterning.[26,83] In general, light ions (i.e., H or He) are utilized for ion-cut of semiconductor wafers because light ion implantation creates narrow damaged layers inside the targets. For the well-known SOI technology,[86,87] H^+ ion implantation is usually employed to form a highly damaged layer at the end of ion range in silicon. The fluence of H^+ ions are typically 4×10^{16}–$1 \times 10^{17}\,cm^{-2}$, which enables the formation of gas bubbles in the implanted layer (particularly under thermal annealing treatment).[86] The associated stress leads to fractures at the corresponding depths and the surface layer of silicon flakes off. Similarly, ion implantation plays a significant role in damage layer formation for LN thin film exfoliation.[26] Since LN contains oxygen, He^+ instead of H^+ ions are used for ion implantation in order to avoid the formation of possible H-O bond during ion implantation process.

The energy of He^+ ions is one of the key parameters for LNOI fabrication because it determines the film thickness.[26] The He^+ ions penetrate into the LN crystal, losing their energies and resting at certain depths inside the crystal. This ion implantation process can be modeled by the software packages SRIM (Stopping and Range of Ions in Matter) based on the Monte Carlo calculation.[88] Figure 2 shows the range profiles of He^+ ions in LN crystal at energies of 200, 300, and 400 keV, respectively. Other parameters such as ion implantation uniformity, ion beam scanning rate, and thermal controlling are also crucial for large-scale wafer fabrication because the quality of LNOI thin films is very sensitive to any inhomogeneous processing of sample in sub micrometer-scale structures.[26] Technically, the crystalline orientation of LN wafer is also crucial for surface processing. LN is a pyroelectric crystal. As a result, under thermal treatment, the generated charges of different crystalline faces may be of significant difference

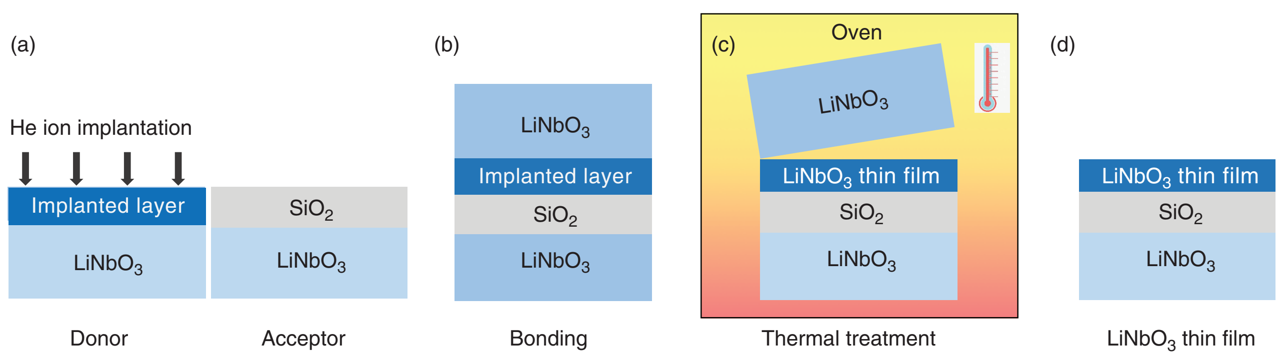

FIG. 1. Fabrication sequence of lithium-niobate-on-insulator thin films by combining ion cut and wafer bonding: (a) preparation of ion-implanted donor wafer and (b) SiO_2-deposited acceptor wafer, (c) wafer bonding, (d) thermal treatment. (e) The final high-quality lithium-niobate-on-insulator thin film after a fine surface polishing process.

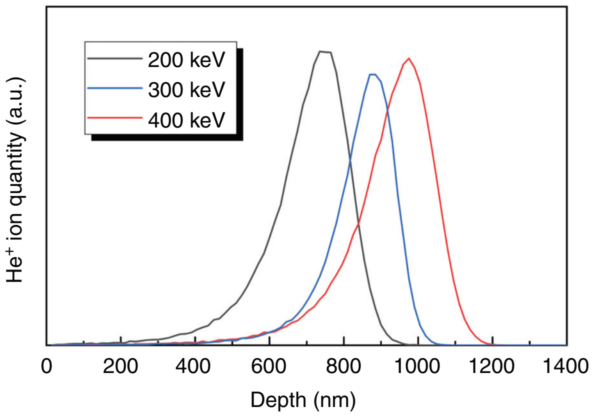

FIG. 2. The range profiles of He ions in $LiNbO_3$ crystal at energies of 200, 300, and 400 keV calculated by the Stopping and Range of Ions in Matter (SRIM) code, respectively.

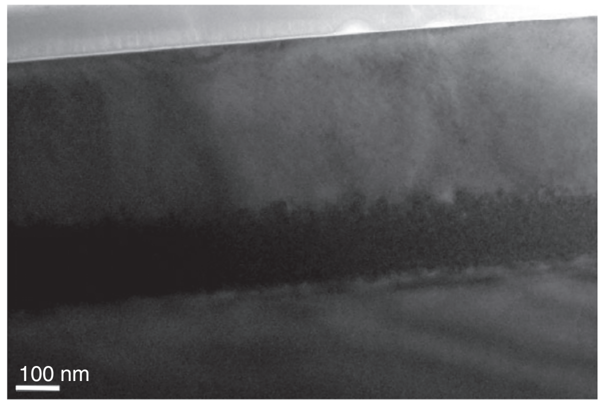

FIG. 3. Transmission electron microscopic (TEM) image for the 120-keV He^+ ion implantation of *x*-cut $LiNbO_3$ at a fluence of $3 \times 10^{16}\,cm^{-2}$.

(e.g., +z or –z face). For large-scale LNOI wafer fabrication, the effect from pyroelectric properties requires removal of the static electric field from charges.

B. Defect engineering

The energy deposition of accelerated He^+ ions onto LN crystalline lattice creates defects or damages at certain regions during implantation process. There are two main effects, namely, the nuclear damage (corresponding to the elastic collisions between ions and target nuclei) and the electronic damage (associated with inelastic collisions between ions and electrons) that are responsible for energetic-ion-induced local crystalline damages and material modification.[6,26,83] Despite the total impact of He^+, ion implantation on the target is a combination of both. For LNOI technology (i.e., in the low-energy regime), nuclear collisions dominate the main defects formation at the end of ion track, while electronic ionization mainly induces point defects in the implantation trajectory regions.[6,26] Defect engineering is generally required to obtain a high-quality LN thin film.

Numerous studies have been performed on the formation and optimization of energetic-ion-induced defects in LN crystal.[83,88–91] For a low-energy He^+ ion implantation, the defect distribution in LN can be divided into three connected zones: a covering layer corresponding to the pre-damaged stage, a heavy damage stage corresponding to the transition layer, and an amorphous layer at the end of ion range.[83,91] Figure 3 shows a typical transmission electron microscopic (TEM) image of the amorphous layer region in a He^+-ion-implanted LN crystal.[83] The formation of He^+ bubbles after thermal annealing treatment at moderate temperature (e.g., around 200 °C) can be observed, resulting in the final splitting of surface layer from bulk.

The pre-damaged layer possesses a large amount of point defects, which leads to a preferential rearrangement of the displaced Nb atoms to vacant octahedral sites. In the transition layer, heavily damaged defect clusters are formed, resulting in a complete random distribution of Nb atoms.[83,91] Figure 4(a) shows the defect concentration as a function of depth for the 325-keV He^+-ion-implanted LN at a fluence of $4 \times 10^{16}\,cm^{-2}$ at a low temperature of 100 K.[83] In addition to the clear evidence of three connected zones, it has been also found that the defect evolution and accumulation is slightly different for *x*- and *z*-cut LN due to the dissimilar crystalline lattices.[83]

For practical applications, LNOI thin films are required to possess no defect. For He^+-ion-implanted LN, the amorphous and transition layer are removed after the CMP step, while the quality of the LNOI thin film depends only on the covering layer with point defects. An efficient approach to annihilate these point defects and to recover crystalline lattice is thermal annealing treatment. A number of studies report the behaviors of damage evolution for ion-implanted LN under thermal annealing.[6,26,83] It has been manifested that, as the temperature elevated up to above 400 °C, the point defects can be significantly removed after thermal annealing for a few hours. This is also confirmed indirectly by measuring the EO coefficient (r_{33}) and the TEM image of the sample [as shown in Fig. 4(b)].[83] The reported r_{33} for the LNOI thin film reaches 99% of that in the bulk, suggesting well-preserved single crystalline features in the thin film.[26,92] This also benefited from the high thermal stability of the SiO_2 cladding. In contrast, for LNOI with a cladding layer of polymer benzocyclobutene (BCB), the thermal treatment cannot be performed at high temperatures because polymers are prone to thermal degradation.[18,26,82,93] As a result, the EO coefficient in the case with BCB layer is reduced by 70% in comparison to the bulk.[18,26] Detailed discussion concerning different bonded materials can be found in Subsection II C.

C. Wafer bonding of LNOI

In general, using LN as the supporting substrate enables direct bonding of exfoliated LN thin films,[27] and this is sufficient for fabrication of micrometer-thick thin films with small diameters (e.g., $< 1\,cm^2$). However, in order to achieve large-size (e.g., up to 4-inch-diameter) LNOI thin films with uniform thicknesses, the wafer bonding process prior to thin film exfoliation is actually one of the most critical and important steps. In order to achieve a high-quality wafer bonding with strong and homogeneous bonding strength distribution, one has to carefully consider the optical flatness, roughness, and cleanness of the bonding surfaces as well as the thermal coefficient mismatch between the supporting substrate and the implanted layer. These factors are all very important for the resultant bonding strength

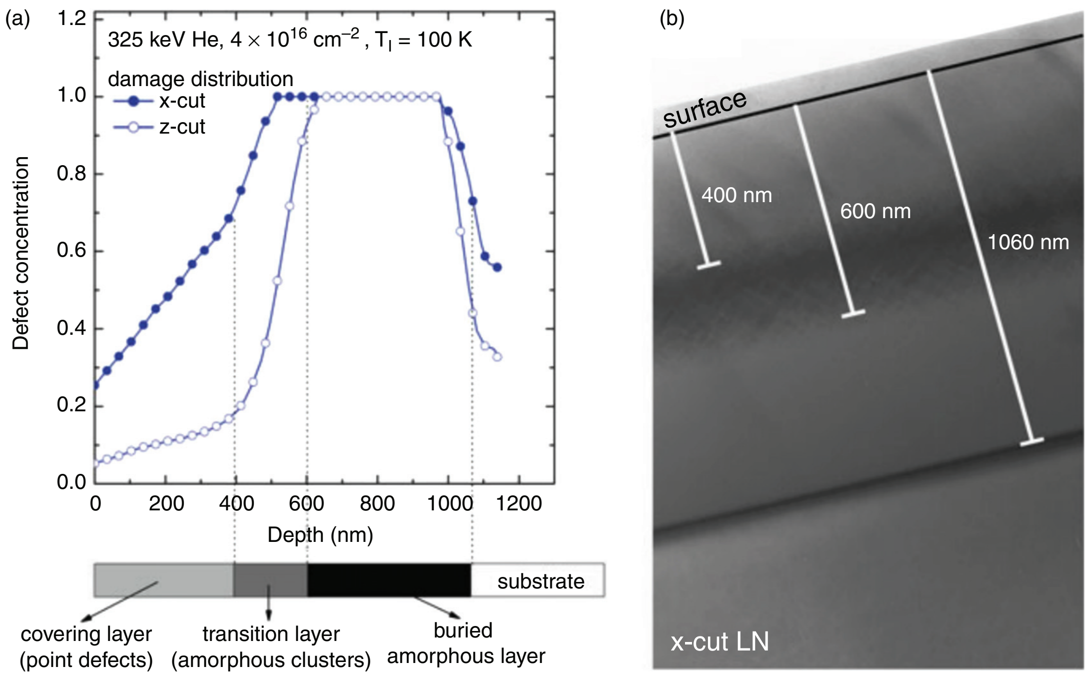

FIG. 4. (a) Defect concentration as a function of depth for the 325-keV He ion implantation of the *x*- and *z*-cut $LiNbO_3$ at a fluence of $4 \times 10^{16}\,cm^{-2}$ at $T = 100\,K$, and (b) transmission electron microscopic (TEM) image of the *x*-cut $LiNbO_3$ sample.[83] Reproduced with permission from W. Wesch and E. Wendler, *Ion Beam Modification of Solids* (Springer, 2016). Copyright Springer International Publishing Switzerland, 2016.

and thus the final LNOI thin film quality, any of them could have a significantly negative impact.[18,26]

In order to achieve a large refractive index difference, generally, a micrometer-thickness layer of low-refractive-index SiO_2 is deposited on the supporting substrate by plasma enhanced chemical vapor deposition (PECVD) before wafer bonding. Therefore, the final configuration is a sandwich structure, namely, LN film/SiO_2 layer/LN substrate.[18,26,92] For EO application, usually, an electrode layer (introduced by metal deposition prior to SiO_2 layer deposition) below the LN thin film is required.[18] Despite the fact that the donor and acceptor wafers can be bonded together at room temperature, a proper thermal treatment is highly desired with a view to achieving high-quality thin film exfoliation. As a result of the ion-induced crystalline lattice damage and the relatively high surface roughness, the overall quality of the as-prepared LNOI thin film is in general relatively poor. In view of this, a well-studied post-processing procedure has been developed, elevating the thin film quality up to the commercial product level.[26] Such an optimized post-processing usually includes a high-temperature thermal annealing treatment (usually at a temperature of 500 °C for 2 h), followed by a fine CMP process. Besides further enhancing the bonding strength, the former step is also beneficial for removing embedded ion-induced point defects and therefore recovering single-crystalline properties of LNOI thin film. Moreover, a surface functionalization procedure prior to wafer bonding has been proven to be very useful for OH-terminated surface formation, leading to a high bonding strength[41] as well as a uniform LN film/SiO_2 layer interface without any residual stress and defects (as shown in Fig. 5). On the basis of the preliminary 5-step fabrication sequences and a series of following

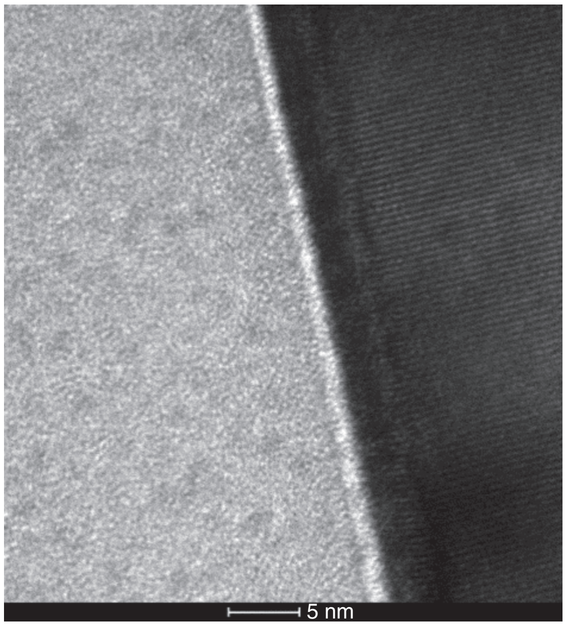

FIG. 5. A transmission electron microscopic (TEM) image of the $LiNbO_3/SiO_2$ interface of a lithium-niobate-on-insulator thin film, indicating the high-quality boundary.

optimization post-processing steps, high-quality, large-size (4 inches in diameter or even larger) LNOI thin films (usually with a thickness from 300 nm to 900 nm) on LN and even on Si substrates have been developed successfully and become commercially available.[94] Using Si as supporting substrate (i.e., LNOSi thin films) makes the ion-cut LNOI technology more compatible with complementary metal-oxide-semiconductor (CMOS) fabrication technology and existing photonic platforms, which has attracted significant interest in the past years because it combines the excellent EO properties of LN and the well-developed Si-based photonics.[16,21,23,25,95] Although, the heterostructure approach usually suffers from reduced overlap between optical fields and relatively weak bonding strength due to the large thermal expansion coefficient mismatch between LN and Si.

Alternatively, following the similar fabrication sequence illustrated in Fig. 1, but replacing the SiO_2 layer with an adhesive polymer BCB layer, LNOI with wafer sizes of several cm^2 can be achieved.[18,26,82,93] In contrast to SiO_2, polymer adhesive is able to significantly relax the strict requirements in respect to optical flatness, roughness, and cleanliness of the bonded surfaces. However, since polymers are prone to thermal degradation, high-temperature (typically $>300\,^{\circ}$C) thermal treatment is generally impossible, which means that one of the key steps in the post-processing procedure for bonding strength enhancement is missing. Nevertheless, a number of good-quality integrated photonic devices have been fabricated based on LN/BCB/LN thin films.[18,26,82,93]

III. FABRICATION OF LNOI ON-CHIP OPTICAL COMPONENTS

A. Micro-/Nano-structuring of LNOI on-chip waveguides

Enabled by the high refractive index contrast between the bonding pairs (i.e., $n_{o,LN} \approx 2.21$, $n_{e,LN} \approx 2.14$, and $n_{SiO2} \approx 1.44$ at 1.55 μm) of LNOI thin film, strong optical confinement and therefore effective fabrication and flexible design of on-chip waveguides and guiding-related photonic devices can be achieved. In particular, with the tremendous development of advanced surface micro-/nano-structuring methods such as photolithography as well as chemical and dry etching techniques, high-density LNOI-based PIC, which consists of different on-chip LN optical components and diverse photonic functionalities, is within reach.[16,23,25,26] To this end, fabrication of low-loss on-chip waveguides with good optical confinement is of significant importance since it is in general the very first step toward the construction of high-performance photonic chips.

Proton exchange, as a conventional and well-established diffusion-based technique for LN waveguide fabrication (generally based on LN bulk), can be directly adopted in fabrication of LNOI on-chip waveguides as well. The desired refractive index contrast in proton-exchanged waveguides is achieved via substituting mobile ions in LN by diffusant ions in surrounding melt solution, forming a planar or an embedded waveguide geometry (with surface masking/patterning). Despite the relatively small refractive index change (usually <0.1 in the in-plane dimension), proton-exchanged on-chip waveguides possess smooth surfaces and boundaries, resulting in reduced waveguide propagation losses (a propagation loss as low as 0.2 dB/cm with a mode size of 3 μm^2 has been experimentally determined in an proton-exchanged embedded waveguide).[96–98] It is also worth mentioning that proton-exchanged on-chip waveguides generally exhibit strong polarization-dependent guiding properties, limiting their applications in, for example, nonlinear optics.

In contrast to the proton-exchanged on-chip embedded waveguides, the unwanted LN on either side of dry-etched on-chip ridge waveguides have been completely removed, offering a very high refractive index contrast (>1 in the in-plane dimension) and thus enabling a high-integration density and tight waveguide bends. A frequently used fabrication sequence of dry-etched on-chip LNOI ridge waveguides is schematically illustrated in Fig. 6. Briefly, the desired surface pattern is first defined by lithography technology [e.g., photolithography or electron-beam lithography (EBL)] and then transferred to the LN thin film layer (usually with a ridge geometry) by inductively coupled plasma reactive ion etching (ICP-RIE) technique.[18,23,26] Following the preliminary fabrication steps, a chemical etching and a subsequent CMP process are highly required with a view to removing the residual mask and smoothing the waveguide surfaces/side walls.[99] Since the surface and sidewall roughness is generally inevitable only by optimizing dry-etching parameters, the final fine polishing step is therefore very essential for dry-etched waveguides in order to obtain low-loss on-chip photonic devices. In pioneering studies, with non-optimized fabrication or fine polishing recipes, the lowest propagation loss of dry-etched on-chip LNOI waveguides was determined to be as high as 10 dB/cm.[100] This value was significantly reduced to be around 3 dB/cm (an on-chip ridge waveguide with an effective mode size of around 0.52 μm^2) by optimizing the dry-etching recipe, exhibiting an efficient second harmonic generation (SHG).[101] Subsequently, by

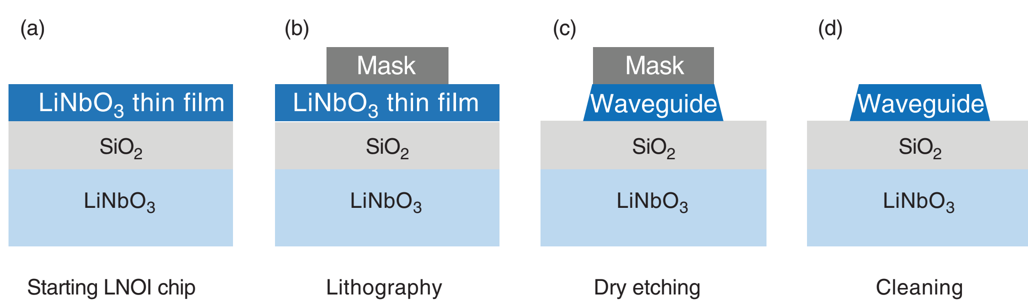

FIG. 6. Fabrication sequence of $LiNbO_3$ on-chip waveguides by employing lithography technology (e.g., photolithography or electron-beam lithography) and subsequent dry-etching approach. A starting $LiNbO_3$ chip is shown in (a). (b) An etching mask is deposited and patterned on the $LiNbO_3$ chip, followed by a dry-etching process to remove unprotected $LiNbO_3$, forming ridge waveguide structures. (d) A post-cleaning process is necessary in order to remove the residual etching mask.

applying a multi-pass exposure of the photoresist (proposed by Lončar's group at Harvard University), dry-etched on-chip waveguides with an extremely low propagation loss of 2.7 dB/m and a bending loss of 9.3 dB/m at telecom wavelengths have been demonstrated, representing by far the state-of-the-art quality for LNOI on-chip ridge waveguides [as shown in Figs. 7(a) and 7(b)].[102] Follow-up work from the same group applied identical fabrication procedure and obtained a propagation loss of 6 dB/m at visible wavelengths.[103]

As for on-chip LNOI ridge waveguides, sidewall roughness is one of the dominating factors that determine the total waveguide loss. Therefore, systematical research on further smoothing the waveguide sidewall is highly desirable and highly significant to improving the overall performance of LNOI photonic chips. Thus far, many attempts have been made to effectively reduce the sidewall roughness either by means of an optimized etching process [resulting in a propagation loss as low as 0.4 dB/cm and a RMS roughness of < 2 nm; an SEM view is shown in Fig. 7(c)][104] or a sidewall fine polishing post-process [resulting in an ultralow propagation loss of 4 dB/m; an SEM view is shown in Fig. 7(d)].[99] In the latter case, the smooth side walls and the scattering-loss reduction (by more than one order of magnitude) of the LNOI on-chip ridge waveguide is achieved by gently polishing the ridge's sidewalls and simultaneously protecting the top surfaces by a metal layer, which was first developed by Buse's group at Freiburg University.[99] Recently, by combining sidewall fine polishing technique and full spectrum laser (fs-laser) ablation for surface patterning [namely, the so-called Chemo-Mechanical Polish Lithography (CMPL) proposed by Cheng's group at East China Normal University],[105] a multi-centimeter-long LNOI on-chip waveguide (with a surface roughness of 0.452 nm) with a propagation loss as low as 0.027 dB/cm has been fabricated.[106] Based on such a fs-laser-assisted CMP fabrication process, subsequently, a reconfigurable multifunctional PIC [combining a high-extinction-ratio beam splitter, a 1×6 optical switch, and a balanced 3×3 interferometer, a photograph of the fabricated chip is shown in Fig. 7(e)][107] and electro-optically switchable optical true delay lines (OTDLs) of meter-scale lengths [a photograph of the fabricated chip is shown in Fig. 7(f)] have been demonstrated,[108] suggesting the great potential of such a hybrid technique. Given this, we believe that a proper combination of an optimized surface patterning/structuring (e.g., EBL+RIE) and a subsequent fine polishing process (e.g., CMPL) is an effective and promising approach toward the realization of absorption-limited LNOI on-chip waveguides, although lots of effort on processing

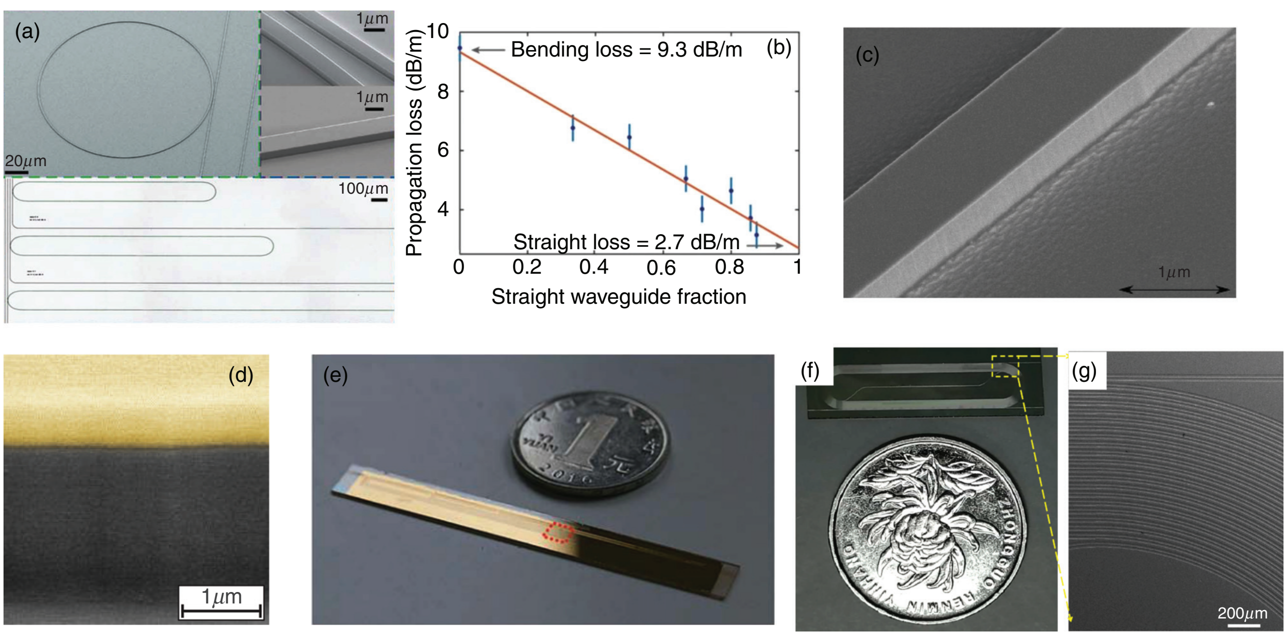

FIG. 7. (a) Scanning electron microscope (top) and optical microscope (bottom) images of LNOI on-chip microring resonators and coupling waveguides with (b) a propagation loss of 2.7 dB/m and a bending loss of 9.3 dB/m at telecom wavelengths.[102] Reproduced with permission from M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Lončar, "Monolithic ultra-high-Q lithium niobate microring resonator," Optica **4**(12), 1536–1537 (2017). Copyright 2017 Optical Society of America. The scanning electron microscope images of $LiNbO_3$ on-chip waveguide side-walls achieved by (c) optimized fabrication process (resulting in a propagation loss as low as 0.4 dB/cm)[104] and (d) sidewall fine polishing post-process (resulting in an ultralow propagation loss of 4 dB/m).[99] Reproduced with permission from I. Krasnokutska, J. J. Tambasco, X. Li, and A. Peruzzo, "Ultra-low loss photonic circuits in lithium niobate on insulator," Opt. Express **26**(2), 897–904 (2018) and R. Wolf, I. Breunig, H. Zappe, and K. Buse, "Scattering-loss reduction of ridge waveguides by sidewall polishing," Opt. Express **26**(16), 19815–19820 (2018), respectively, Copyright 2018 Optical Society of America. (e-g) Digital camera pictures of the LNOI chips (fabricated by the fs-laser-assisted CMP process, with a propagation loss of 2.7 dB/m) placed nearby 1 Chinese yuan (CNY) coin, demonstrating (e) a reconfigurable multifunctional PIC[107] and (f, g) an electro-optically switchable optical true delay line.[108] Reproduced with permission from R. Wu, J. Lin, M. Wang, Z. Fang, W. Chu, J. Zhang, J. Zhou, and Y. Cheng, "Fabrication of a multifunctional photonic integrated chip on lithium niobate on insulator using femtosecond laser-assisted chemomechanical polish," Opt. Lett. **44**(19), 4698–4701 (2019), Copyright 2019 Optical Society of America, and J.-X. Zhou, R.-H. Gao, J. Lin, M. Wang, W. Chu, W.-B. Li, D.-F. Yin, L. Deng, Z.-W. Fang, J.-H. Zhang, R.-B. Wu, and Y. Cheng, "Electro-optically switchable optical true delay lines of meter-scale lengths fabricated on lithium niobate on insulator using photolithography assisted chemo-mechanical etching," Chin. Phys. Lett. **37**(8), 084201 (2020), Copyright 2020 Chinese Physical Society and IOP Publishing Ltd.

parameter optimization is required. Nevertheless, based on the breakthroughs and achievements on waveguide quality improvement, a number of additional photonic functionalities and nanostructures, e.g., Bragg gratings,[109,110] optical metasurafaces,[111,112] quantum dots,[113] and photodetectors,[114] integrated on waveguides have been demonstrated, opening up many exciting possibilities and opportunities for further enriching the overall integration level and on-chip multi-functionalities of LNOI-based PICs. Besides sidewall roughness, dry-etched LNOI on-chip waveguides usually suffer from significant sidewall angle (generally $>25°$ with respect to the vertical direction), which has to be well considered while designing.[16,23,25]

Precise diamond blade dicing is a more straightforward approach for definition of almost vertical waveguide sidewalls,[115] which may have the potential to overcome the large-angle sidewall issue that occurred in dry-etched on-chip waveguides. By using this method, LN/PPLN on-chip ridge waveguides with a propagation loss as low as 0.3 dB/cm has been demonstrated.[116,117] It has been found that a better sidewall verticality can be obtained by applying a relatively deep cut, but the fabricated ridge waveguide is prone to cracking.[116] A significant shortcoming of precise diamond blade dicing, as a purely mechanical method, for definition of on-chip waveguides is that it lacks the ability of fabricating bended geometries, greatly limiting its applications in construction of PICs.

B. LNOI on-chip whispering-gallery resonators

Whispering-gallery-mode-based microresonators, also called whispering-gallery resonators (WGRs), usually in the geometrical form of microdisk or microring, are compact optical devices that have a variety of applications in photonics.[118,119] A WGR is able to offer a remarkably strong optical confinement and an enhanced intracavity optical intensity even without any optical coatings, which is very intriguing for wavelength-sensitive applications such as wavelength filters, nonlinear frequency converters, lasers, and sensors.[118,119] In this respect, chip-integrated WGRs are highly promising for multifunctional PICs and lab-on-a-chip platforms.[18,23]

Following the above-mentioned on-chip waveguide processing technology, dry etching was first employed for the fabrication of on-chip LN microresonators.[82,93] The fabricated microring (first demonstrated by Günter's group at ETH Zurich) has a large radius of 100 μm but exhibits a relatively low Q factor of only 4×10^3 at 1.55 μm, which can be attributed to the un-optimized etching process and thus high surface/sidewall roughness without any fine polishing treatment. In view of this, since then, a lot of effort has been made on improving the quality of LNOI on-chip WGRs, approaching step by step the theoretical upper limit of Q-factors (can be $>10^8$ at telecom wavelengths in the case of a LN microresonator).[16,18,23,25,26] For example, by employing laser lithography technique in a following study from the same group, an on-chip microring (with a radius of 20 μm) exhibiting a Q-factor of 1×10^4 at telecom wavelengths was fabricated.[18] By applying thermal treatment (with a temperature up to LN's melting point) of an as-fabricated 20-μm-diameter on-chip microdisk, the surface can be further smoothened via surface tension reshaping, resulting in an improved Q-factor of 2.6×10^4 at telecom wavelenths.[120] Based on the heterogeneous approach (demonstrated by Fathpour's group at the University of Central Florida), a Q-factor of 7.2×10^4 at telecom wavelengths has been determined in a 300-μm-diameter on-chip microring.[95] In 2014, three separate studies demonstrated LNOI on-chip microresonators with Q-factors exceeding 10^5 by using fs-laser ablation followed by focused ion beam (FIB) milling (first proposed by Cheng's group at East China Normal University)[121] and the above-mentioned dry-etching approach,[122,123] respectively. The achieved Q-factors are 2.5×10^5 (by using fs-laser ablation followed by FIB milling),[121] 4.84×10^5 (by using dry etching),[122] and 1.02×10^5 (by using dry etching),[123] respectively (the SEM views of these high-quality on-chip resonators are shown in Fig. 8). From then on, these two fabrication recipes have matured gradually and become the dominant techniques in demonstrating high-Q-factor LNOI on-chip microresonators, as implied by the most recent publications as well.[16,23,25,26] By further combining with optimized nano-fabrication/structuring technologies and post-processing methods such as CMP and sidewall polishing, on-chip microresonators exhibiting Q-factors $>10^6$ (the state-of-the-art Q-factor is at 10^7 level) at both telecom and visible wavelengths have been obtained,[99,102,103,124,125] demonstrating a large variety of important on-chip photonic applications as well as some interesting photonic phenomena.[16,23,25,26]

C. LNOI on-chip photonic-crystal-based structures

PhC-based structures (generally made of periodic dielectrics with different refractive indices) are designed to form the energy band structure for photons and thus to influence the motion of photons with specific frequencies, which has some similarities to that of semiconductor crystals for defining allowed and forbidden electronic

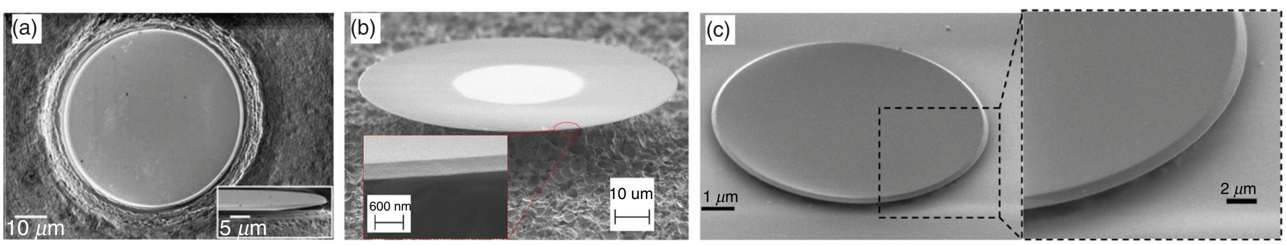

FIG. 8. The scanning electron microscopic images of the first demonstrations of $LiNbO_3$ on-chip microdisk resonators with $Q > 10^5$ (at telecom wavelengths) fabricated by (a) fs-laser microfabrication[121] or (b, c) dry etching,[122,123] respectively. Reproduced with permission from J. Lin, Y. Xu, Z. Fang, M. Wang, N. Wang, L. Qiao, W. Fang, and Y. Cheng, "Second harmonic generation in a high-Q lithium niobate microresonator fabricated by femtosecond laser micromachining," Sci. China-Phys. Mech. Astron. **58**(11), 114209 (2015), Copyright 2015 Science China Press and Springer-Verlag Berlin Heidelberg; R. Wang and S. A. Bhave, "Free-standing high quality factor thin-film lithium niobate microphotonic disk resonators," ArXiv: 1409.6351 (2014), Copyright 2014 The Author(s); and C. Wang, M. J. Burek, Z. Lin, H. A. Atikian, V. Venkataraman, I. C. Huang, P. Stark, and M. Lončar, "Integrated high quality factor lithium niobate microdisk resonators," Opt. Express **22**(25), 30924–30933 (2014), Copyright 2014 Optical Society of America.

energy bands.[126] The unique property of PhC-based structures enables flexible control of light flow and produces effects that are impossible with conventional optics, which are very promising and of significant importance for enriching photonic functionalities of PICs based on LNOI platform.

In order to define the desired dielectric periodicity based on LNOI platform, the pioneering studies utilized FIB milling, which has been proven to be a simple and effective approach for fast prototyping of LNOI-based PhC structures, including 1D PhC nanobeam cavities and 2D PhC slabs. As a result, an on-chip 2D PhC slab (exhibiting a photonic bandgap in the telecom wavelength range as well as an extinction ratio of up to 20 dB)[127,128] and an on-chip 1D PhC nanobeam micro-cavity (with a Q-factor of 156 and an extinction ratio of 12.5 dB at telecom wavelengths) embedded on an on-chip ridge waveguide[129] have been successively demonstrated. Alternatively, by further combining with the so-called ion beam enhanced etching (IBEE) technique,[130] FIB milling of PhC-based L3 cavities (with a Q-factor of 500–600 in the telecom wavelength range) has been fabricated in self-suspended LN membranes, allowing for effective SHG.[131–133] Similar to the ion-cut technology, IBEE also relies on the ion-induced localized lattice damages, resulting in a highly damaged region where the physical stability against chemical etching is strongly degraded, enabling submicrometric surface structuring/patterning.[6] In these above-mentioned studies, the optical Q-factors of the fabricated on-chip PhC structure are all $<10^3$, which strongly limits their applications for fabrication of functional LNOI-based photonic devices. In view of this, by employing the optimized "EBL + dry etching" approach (as discussed in Sec. III A), recently, the Q-factors of LNOI PhC-based structures [including 1D on-chip PhC nanobeam cavities and 2D on-chip PhC slabs; SEM views are shown in Figs. 9(a), 9(c), and 9(d)] have been upgraded by more than two orders of magnitude, reaching as high as $>10^5$,[50,59,134] and most recently 1.41×10^6 [a simulated optical mode profile of which is shown in Fig. 9(b)][51] over a broad telecom band. The latter represents the state-of-the-art Q-factor for LNOI on-chip PhC-based structures.

IV. TECHNOLOGICAL ADVANCES

A. Surface domain engineering and periodically poled LNOI (PPLNOI) technology

Due to its ferroelectric characteristics, quasi-phase matching (QPM) can be realized in LN by creating periodic inversion of ferroelectric domains.[1] Such domain patterns in LNOI thin films can be defined in two different ways: either the thin film is exfoliated from an already poled bulk crystal prior to wafer bonding or the thin film is poled afterward. In the past decade, tremendous progress has been made toward high-quality PPLNOI thin film platform for on-chip photonic devices along both of these two paths, which will be briefly described in the following.

Initial attempts in preparing PPLNOI thin films focuses on fabrication of a freestanding PPLN thin film and then bonded it onto an insulator substrate.[135] In spite of the large thickness (10 μm) of the final thin film therein, the domain periodicity and bulk-like nonlinearity were found to be well preserved during ion beam processing. Further thinning of PPLNOI thin films to <1 μm level is reachable by applying optimized processing strategy.[18] This "poled slice transfer" approach may require additional efforts on statistically studying wafer bonding or thinning technologies, which could be quite challenging due to the periodically distributed domain structures, but it ensures the high quality of domain periodicity once the PPLN thin film survives the final thinning and fine polishing procedures. Particularly, this approach is compatible with both x- and z-cut LNOI platforms, though in most cases, the latter is more favorable since it always provides access to the largest second-order nonlinear optical ($d_{33} \approx 27$ pm/V) and electro-optic coefficients ($r_{33} \approx 30$ pm/V) of LN. Along this path, based on z-cut PPLN thin-film waveguides (with thicknesses of 700 nm and losses as low as 0.7 dB/cm), on-chip tunable broadband nonlinear frequency converters [sum frequency generation (SFG) and SHG with their respective bandwidths of 15.5 nm[136] and 2 THz[137] at 1530-nm wavelength, by simultaneously satisfying QPM and group-velocity matching conditions] have been fabricated by combining crystal ion slicing and wafer bonding. The broad bandwidth, in addition, can be largely modulated from the mid-infrared (MIR) region to the telecommunications band by engineering the PPLN thin film thickness,[137] i.e., control of the generated central wavelength of nonlinear frequency conversion. An example of PPLNOI thin film for nonlinear optical frequency conversion is shown in Fig. 10(a). By further combining with precise diamond blade dicing, high-quality PPLNOI ridge waveguides (with losses as low as 0.3 dB/cm) were fabricated, demonstrating electrically controllable active mode converters[117] and nonlinear frequency converters, as schematically illustrated in Fig. 10(b).[138,139] In the latter case, the integration of cascading processes of EO coupling and SHG/SFG in the same PPLNOI

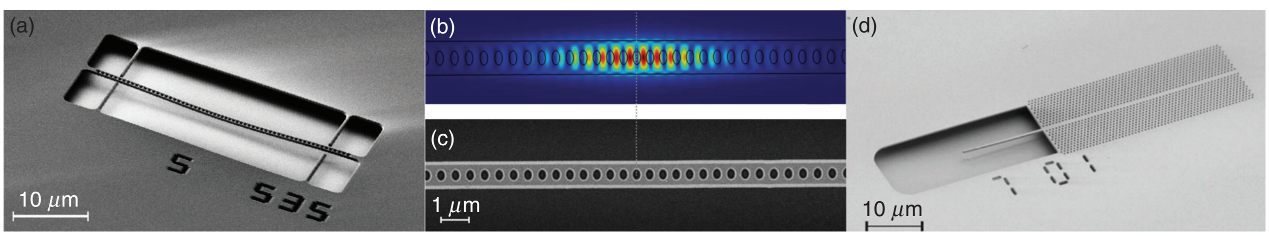

FIG. 9. Scanning electron microscopic images of $LiNbO_3$ on-chip one-dimensional photonic-crystal nanobeam cavities [with an optical Q of (a) 1.09×10^5,[134] and (c) 1.41×10^6,[51] respectively] and (d) 2D photonic-crystal slab (with an optical Q of 3.51×10^5) fabricated by the "electron-beam lithography + dry-etching" approach.[50] (b) The simulated optical mode field profile of the fabricated $LiNbO_3$ on-chip one-dimensional photonic-crystal nanobeam cavity by Li *et al.*[51] Reproduced with permission from H. Liang, R. Luo, Y. He, H. Jiang, and Q. Lin, "High-quality lithium niobate photonic crystal nanocavities," Optica **4**(10), 1251–1258 (2017), Copyright 2017 Optical Society of America; M. Li, H. Liang, R. Luo, Y. He, J. Ling, and Q. Lin, "Photon-level tuning of photonic nanocavities," Optica **6**(7), 860–863 (2019), Copyright 2019 Optical Society of America; and M. Li, H. Liang, R. Luo, Y. He, and Q. Lin, "High-Q 2D lithium niobate photonic crystal slab nanoresonators," Laser Photonics Rev. **13**(5), 1800228 (2019), Copyright 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

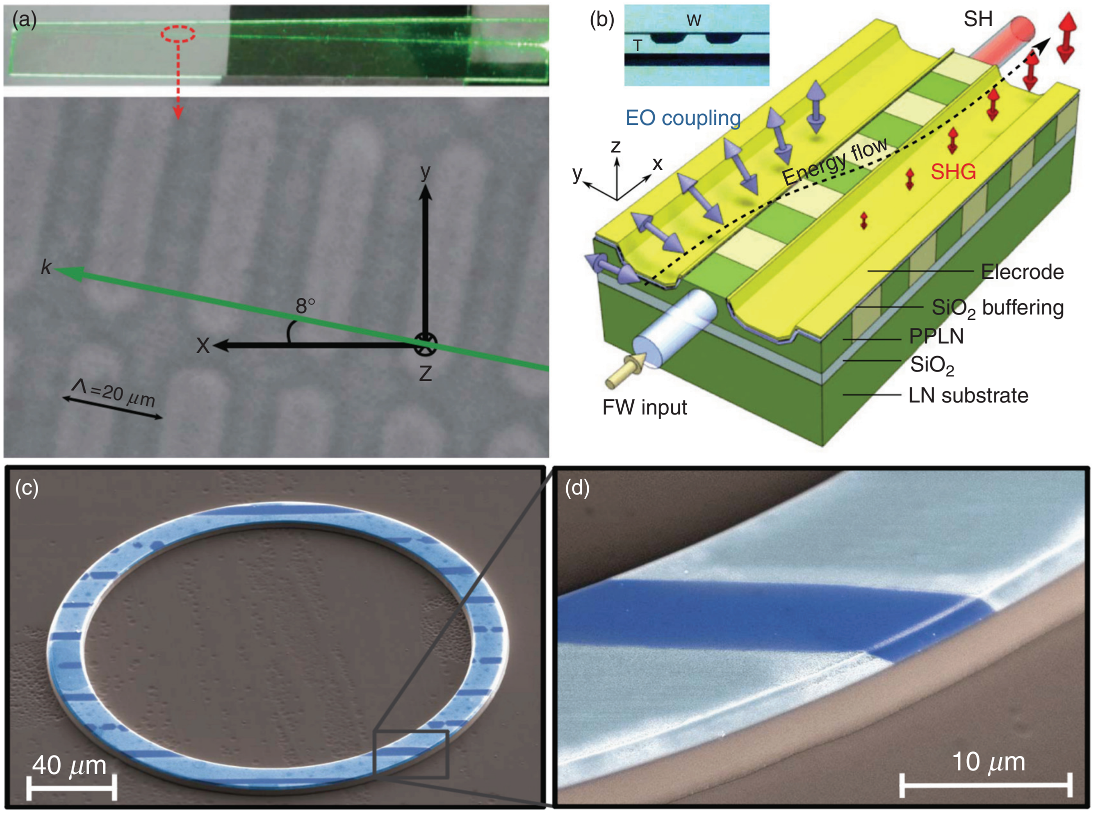

FIG. 10. (a) The observed light confinement and top view of the periodic quasi-phase-matched structure of a periodically poled-lithium-niobate thin film.[137] Reproduced with permission from L. Ge, Y. Chen, H. Jiang, G. Li, B. Zhu, Y. a Liu, and X. Chen, "Broadband quasi-phase matching in a MgO:PPLN thin film," Photonics Res. 6(10), 954–958 (2018), Copyright 2018 Chinese Laser Press. (b) Schematic of the cascading electro-optic coupling and second harmonic generation process in the periodically poled-lithium-niobate ridge waveguide.[138] The arrows indicate the polarization direction of the optical field. Reproduced with permission from T. Ding, Y. Zheng, and X. Chen, "Integration of cascaded electro-optic and nonlinear processes on a lithium niobate on insulator chip," Opt. Lett. **44**(6), 1524–1527 (2019), Copyright 2019 Optical Society of America. (c) False-color scanning electron microscopic images of the periodically poled-lithium-niobate on-chip microring resonator WGR (with an intrinsic *Q*-factor of 1.3×10^6) and (d) zoomed view.[41] Reproduced with permission from R. Wolf, Y. Jia, S. Bonaus, C. S. Werner, S. J. Herr, I. Breunig, K. Buse, and H. Zappe, "Quasi-phase-matched nonlinear optical frequency conversion in on-chip whispering galleries," Optica **5**(7), 872–875 (2018), Copyright 2018 Optical Society of America.

ridge waveguide, enables efficient modulation of the output optical field, i.e., control of the generated light intensity of nonlinear frequency conversion. These above-mentioned results indicate that the integration of EO effect and nonlinear wave mixing can provide a way to manipulate photonic conversion on LNOI platform in an electrically controllable manner. In a separate study, Buse's group at the University of Freiburg fabricated the first on-chip *z*-cut PPLNOI microring resonator [with an intrinsic *Q*-factor of 1.3×10^6, the SEM views of which are shown in Figs. 10(c) and 10(d)] and demonstrated SHG with a normalized conversion efficiency of 9×10^{-4} mW^{-1}. The PPLNOI thin film (with thickness of 2 μm) therein was prepared by wafer bonding, lapping, and polishing, and subsequently structured by conventional semiconductor manufacturing techniques.[41]

In order to fully exploit the largest $\chi^{(2)}$ tensor element d_{33}, the electric field for periodic poling of LN is often applied along *z* axis. This means that the whole design and operation strategy for direct poling of *z*- and *x*-cut LNOI thin films is technically very different. Typically, direct poling of *z*-cut LNOI thin films requires electrodes below the active layer. This method is well understood, yet it is not fully applicable for LNOI because its intermediate insulating SiO_2 layer prevents/impedes electrical current flow and, hence, high-quality domain growth. Moreover, it may also be disadvantageous for waveguiding, which leads to additional damping if the electrode is directly bonded/deposited beneath the active layer. To achieve direct poling of *x*-cut thin films, in-plane distributed electrode designs/patterns deployed on the top surface of the thin film are sufficient, i.e., surface poling geometry. Therefore, from a technological point of view, the *x*-cut geometry is more favorable due to the possibility that the entire process can be performed on the film surface. In addition, surface poling features good compatibility to standard top-down lithography and waveguide fabrication processes as well as high flexibility in electrode configurations, rendering it thus far the most commonly used and well-studied direct poling technique. However, surface poling is also challenging; many different parameters, such as electrode tip shape, poling pulse parameters, and nucleation pulses, during the poling process have to be well studied since even a slight change could significantly affect the performance and homogeneity of the final result.[140]

For a considerable period of time, experimental demonstration of PPLN on-chip photonic devices by direct poling was missing, which can be partly attributed to the non-optimized film quality and the challenges in applying electric field effectively, despite the fact that direct writing of domain patterns can be realized even by employing AFM-tip voltages (UDC $\leq$ 50 V).[141,142] In recent years, in contrast to that of *z*-cut, there has been a significant effort toward direct poling of *x*-cut PPLNOI thin films, in which TE-polarized modes are usually excited to make use of the highest second-order nonlinear tensor component d_{33} for efficient parametric frequency conversion. The first direct poling of PPLNOI thin film for SHG was realized based on an *x*-cut LNOI platform, on top of which comb-shaped electrodes were patterned while HV pulse poling was applied, demonstrating effective periodic domain inversion verified by confocal SH analysis [as schematically illustrated in Fig. 11(a)].[143] By using the similar poling strategy together with subsequent heterogeneous integration (by rib-loading the LN thin films with SiN strips), Bowers's group at the University of California demonstrated the first on-chip *x*-cut PPLN waveguide-based nonlinear frequency converter, featuring a SHG conversion efficiency of 160%$\text{W}^{-1}\text{cm}^{-2}$.[144] This value is then elevated by more than an order of magnitude (up to 2200% and 2600%$\text{W}^{-1}\text{cm}^{-2}$) for SHG of 1.5-$\mu$m radiation, benefiting from low waveguide losses, large mode overlapping, and high-quality periodic poling [an example of the PPLNOI on-chip ridge waveguide is shown in Fig. 11(b)].[46,145] By monitoring the *in situ* SHG power during the periodic poling process and using it as feedback for optimization of the periodic poling parameters in a subsequent study, ultrahigh SHG conversion efficiencies up to 3061% and 4600%$\text{W}^{-1}\text{cm}^{-2}$ have been achieved in on-chip *x*-cut PPLN ridge waveguides.[146,147] By further combining with dispersion engineering via well-designed waveguide geometry, *x*-cut PPLNOI on-chip ridge waveguides producing coherent multi-octave supercontinua are demonstrated [see Fig. 11(c)], which are comprised by multiple spectrally broadened harmonics with reduced pulse energy requirements.[148] Besides waveguides, chip-integrated PPLNOI whispering gallery resonators are highly promising for high-efficiency and compact nonlinear frequency conversion devices. By employing such photonic structures, cavity-enhanced-SHG with a normalized conversion efficiency of 230,000%/W has been demonstrated in an *x*-cut PPLNOI racetrack resonator (with a *Q*-factor of 3.7×10^5) as shown in Fig. 11(d), representing that the nonlinearity in the device has been elevated to the single-photon level.[149]

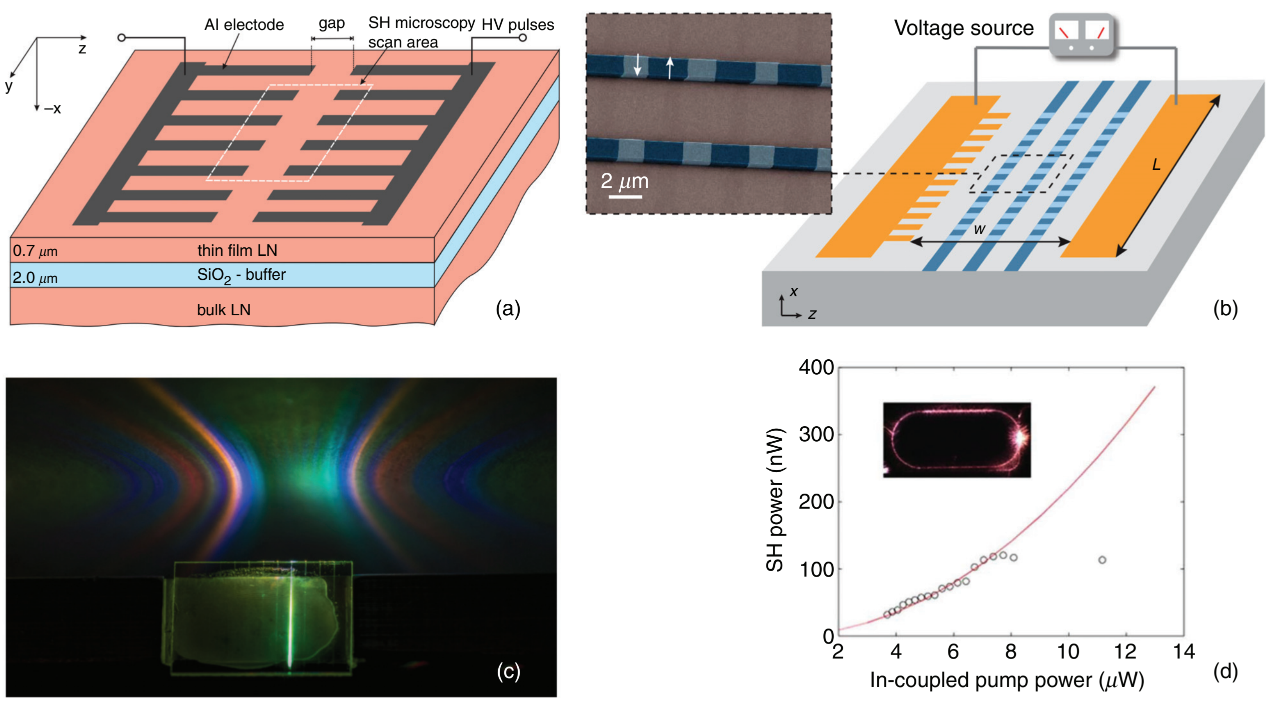

FIG. 11. Schematic of the typical geometry for surface poling of *x*-cut $LiNbO_3$ thin films for achieving (a) thin film[143] and (b) on-chip ridge[46] waveguides as well as second harmonic generation. Reproduced from P. Mackwitz, M. Rüsing, G. Berth, A. Widhalm, K. Müller, and A. Zrenner, "Periodic domain inversion in x-cut single-crystal lithium niobate thin film," Appl. Phys. Lett. **108**(15), 152902 (2016), with the permission of AIP Publishing, and R. Luo, Y. He, H. Liang, M. Li, and Q. Lin, "Highly tunable efficient second-harmonic generation in a lithium niobate nanophotonic waveguide," Optica **5**(11), 1438–1441 (2018), Copyright 2018 Optical Society of America. (c) Photograph of supercontinuum produced from an *x*-cut periodically poled-lithium-niobate on-chip ridge waveguide whose dimension is well-designed for dispersion engineering.[148] Reproduced with permission from M. Jankowski, C. Langrock, B. Desiatov, A. Marandi, C. Wang, M. Zhang, C. R. Phillips, M. Lončar, and M. M. Fejer, "Ultrabroadband nonlinear optics in nanophotonic periodically poled lithium niobate waveguides," Optica **7**(1), 40–46 (2020), Copyright 2020 Optical Society of America. (d) Pump power dependency of the generated second harmonic power from an *x*-cut periodically poled-lithium-niobate racetrack resonator.[149] Reproduced with permission from J.-Y. Chen, Z.-H. Ma, Y. M. Sua, Z. Li, C. Tang, and Y.-P. Huang, "Ultra-efficient frequency conversion in quasi-phase-matched lithium niobate microrings," Optica **6**(9), 1244–1245 (2019), Copyright 2019 Optical Society of America. The inset shows a microscope image of the scattered second harmonic light.

B. Integration with active ions

Many important photonic functionalities/devices, such as compact solid-state lasers and modulators, as well as on-chip optical quantum memories for classical and quantum optical applications, are enabled by various active rare-earth (RE) ions (solid-state emitters), including Er^{3+}, Tm^{3+}, and Pr^{3+}. Particularly, incorporating RE ions into integrated photonic platform enriches the system with new applications in both optoelectronics and quantum information processing.[150] To this end, a single-crystalline thin film platform that is compatible with both planar fabrication of integrated photonic devices and high-quality active ion doping is highly desired. In contrast to the most commonly used silicon for integrated photonics, LN itself is an efficient gain medium and can be doped by different RE ions, rendering it very promising in terms of applications in integrated photonics (LNOI technology) and host materials for active ion doping as well as the efficient combination of both.[151,152] In fact, both RE-doped LN bulks and waveguides have been employed to realize lasers and quantum memories spanning a vast range of wavelengths from the infrared to the telecom band.[153–155] In recent years, the tremendous progress in LNOI technology and advanced nano-structuring techniques raises interest on the research of doping strategies for efficient incorporation of active RE ions and LN thin films, namely RE:LNOI thin films. Since RE:LN crystals are compatible with the standard ion-cut LNOI technology, the most straightforward approach for preparation of high-quality RE:LNOI thin films is applying ion-cut and wafer bonding techniques on a RE:LN wafer instead of pure LN. Although, systematical studies are very much needed to investigate whether the desirable optical properties of the RE ions can be well preserved after significant processing involved in ion-cut and subsequent structuring of the thin film. In particular, when combining active RE ions and in particular, designed resonant structures, such as WGRs and PhC cavities, the target optical properties of the doped ions can be significantly enhanced.

The first RE:LNOI platform is achieved by pre-indiffusion of a thin Er layer into a *z*-cut LN wafer at high temperatures prior to ion-cut and wafer bonding. Based on such a platform, ridge waveguides have been fabricated by subsequent precise diamond dicing, showing identical Raman and fluorescence emissions to that of the bulk materials.[151] Recently, an integrated photonic platform based on *x*-cut Tm:LNOI thin film (300-nm thickness) is demonstrated, in which the Tm^{3+} ions in the thin film exhibit optical lifetimes identical to those measured in bulk crystals.[156] Furthermore, an on-chip waveguide along with a grating coupler are fabricated by using EBL and dry etching processes, showing strong optical absorption and narrow spectral hole burning through transmission measurements. Alternatively, in more recent studies, Er^{3+} and Yb^{3+} ions are doped into *z*-cut LNOI thin film (600-nm thickness) systems by ion implantation technique, respectively.[157,158] Prior to the ion implantation, EBL and dry etching techniques are applied for fabrication of on-chip waveguide and microring structures. After post-implantation annealing treatment, the lattice damage from implantation can be recovered to some extent, and the *Q*-factor of the microring resonator can be elevated up to $>10^6$. In the Er^{3+}-doped case, the extracted transition linewidth and photoluminescence (PL) lifetime of the Er^{3+} ions from optical waveguides are measured to be comparable to those of the Er:LN bulk crystals. Likewise, the non-resonant PL lifetime agrees well with that of the bulk crystals in the case of Yb^{3+} ions doping. While when the microring is pumped on resonant, the Yb^{3+} PL lifetime is slightly shortened, indicating increased light-atom interactions in the cavity. Such an effect is attributed to Purcell-enhanced emissions. Most recently, on-chip microdisk C-band lasers operating at 1530 and 1560 nm wavelengths have been realized based on the Er:LNOI platform.[159–163] The thin film Er:LNOI is prepared by using the standard ion-cut LNOI technology procedure identical to that in Fig. 1, only employing Er-doped LN (with Er^{3+} concentration of 1% mol) instead of pure LN as the donor wafer. The on-chip Er:LNOI microdisk resonators are fabricated either by fs-laser microfabrication followed by chemical etching or FIB milling, followed by buffered oxide etching (BOE), enabling a *Q*-factor as high as 1.02×10^6 and thus a low-pumping-threshold (as low as 250 μW) laser emission.[159,160] Particularly, it has been found that the generated laser wavelengths can be tuned smoothly by varying the pump power, which is considered to be a result of significant photorefractive and thermo-optic effects.[159,160] Figures 12(a) and 12(b) show the spectra of the generated laser and visible emission from Er:LNOI on-chip microdisk resonators. These demonstrated micro-lasers are expected to play important roles in the multifunctional LNOI photonic platform.

Thus far, demonstrations of high-quality RE:LNOI thin films are proven to be compatible with scalable planar fabrication and capable of well preserving the desired optical properties of the active doped ions, albeit post-annealing treatment is usually required for recovering the lattice damage induced by ion implantation. In general, ion implantation and ion indiffusion are both effective approaches for various ion doping, which can also be used for direct patterning embedded photonic devices on the top surface of LNOI. However, the ion-implanted/indiffused photonic devices, e.g., optical waveguides, usually possess very low index contrast (usually <0.1) between core and cladding.[6,16,18,25,26] Apart from the already reported Tm^{3+}, Er^{3+}, and Yb^{3+} ions, many other RE ions can also be efficiently incorporated into LN host material, such as Nd^{3+}, Pr^{3+}, Ho^{3+}, and so on, enabling active optical functionality over a broad wavelength range.[151] The incorporation of RE ions into LN crystals is compatible to the standard ion-cut LNOI technology, thus the mass production of high-quality RE:LNOI thin films is going to be pervasive and we envision more sophisticated on-chip photonic devices will soon appear in integrated photonics. Besides, other types of ion based material-modification technologies, such as ion implantation, ion milling, and ion diffusion, can be also employed to further enrich the integration strategies and functional flexibility of the RE:LNOI platform.

C. LNOI-based heterogeneous integrated systems

In the past decade several technological advances, such as the development of bonding technologies and the deposition/growth of high-quality materials, have facilitated the consideration of material integration as a feasible pathway toward the realization of high-performance heterogeneous PICs.[21,78,79] The achieved breakthroughs have enabled the wafer-scale integration of various materials onto heterogeneous substrates, usually onto CMOS-processing-compatible materials, increasing the quantity as well as the quality of the photonic functionalities that integrated on a monolithic chip. Although most of the advances originate from the more developed Si photonics, they will continue to bring significant impacts to integration photonics platforms based on materials beyond Si, including of course LN.[78] Apart from expanding the device functionality by adding new materials, hybrid LNOI photonics also aims to take advantage of a number

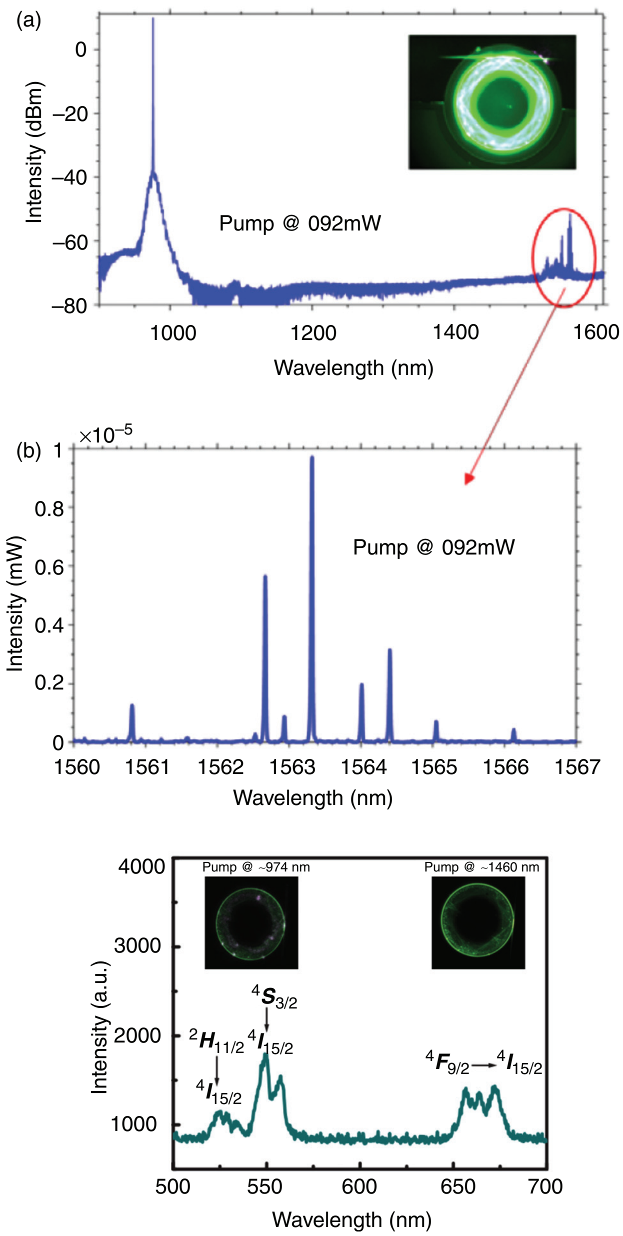

FIG. 12. (a) The generated laser spectrum of the Er:$LiNbO_3$ on-chip microcavity laser with the 976-nm pumping; the inset displays the strong green up-conversion fluorescence.[159] (b) The enlarged spectrum around 1560 nm shows multimode lasing spectrum. Reproduced with permission from Z. Wang, Z. Fang, Z. Liu, W. Chu, Y. Zhou, J. Zhang, R. Wu, M. Wang, T. Lu, and Y. Cheng, "An on-chip tunable micro-disk laser fabricated on Er3+ doped lithium niobate on insulator (LNOI)," ArXiv: 2009.08953 (2020), Copyright 2020 The Author(s). (b) The spectrum of the generated visible emission produced by cooperation up-conversion (CUC) and excited-state absorption (ESA) from an Er:$LiNbO_3$ on-chip microdisk when pumping at 974-nm wavelength; the two insets are images of the visible emission with 974 and 1460 nm laser pumping, respectively.[160] Reproduced with permission from Y. Liu, X. Yan, J. Wu, B. Zhu, Y. Chen, and X. Chen, "On-chip erbium-doped lithium niobate microcavity laser," Sci. China-Phys. Mech. Astron. **64**, 234262 (2021), Copyright 2020 Science China Press and Springer-Verlag GmbH Germany, part of Springer Nature.

of more mature technologies, such as plasmonics, to harness and enhance the interaction of light and hybridized matter at the micro- and nano-scale. In this section, we will introduce the technological advances of the hybrid combination of LNOI with heterogeneous materials and the associated emerging new functions and applications for integrated optics.

1. Thin film LN bonded onto Si photonics platform

The original thought, as it is today, of using LN as one of the materials for heterogeneous integration is to compensate for the absence of optical functionalities provided by second-order nonlinearity ($\chi^{(2)}$), such as Pockels effect and second-order nonlinear frequency conversion, in Si and the standard SOI photonic devices.[86] In early studies, thin-layer LN prepared by ion implantation and thermal treatment were bonded/transferred onto the top of SOI photonic devices.[164–166] In such a hybrid system, the thin film LN serves as a portion of the top cladding of an optical waveguide mode. By combining the EO functionality of LN with the high-index contrast of SOI waveguides, functionalities such as EO tuning/modulation (electro-optic modulator [EOM]) and electric field sensing have been realized in Si microring resonators (with Q-factors $>10^5$).[164–166] In a subsequent study, in particular, hybrid silicon and LN microring resonator EOMs operating at gigahertz frequencies are demonstrated.[167] However, compared to traveling-wave modulators, such as waveguide Mach–Zehnder modulators (MZMs), microring resonators possess fundamentally limited EO bandwidth. Therefore, more recently, based on a foundry-fabricated Si photonics platform bonded to a LN thin film while using photolithography and wafer-scale fabrication, a small-signal electrical-to-optical 3-dB bandwidth exceeding 100 GHz are demonstrated in Si waveguide MZMs (as schematically shown in Fig. 13),[168,169] which is much larger than that of pure Si photonic EOMs (generally limited to 50 GHz).

However, the optical modes supported by such a design are distributed in both the LN and the underlying silicon waveguide (i.e., only the evanescent tail of the optical mode of Si waveguides overlaps the LN cladding layer), which compromises the EO performance, showing a trade-off between high EO bandwidth and low operation voltage. To overcome this issue, two layers of hybrid integrated waveguides (top: dry-etched x-cut LN waveguides, bottom: SOI circuit) and vertical adiabatic couplers (formed by Si inverse tapers and superimposed LN waveguides, serving as interfaces to transfer the optical power between the two layers) have been proposed and employed in hybrid Si/LN waveguide MZMs (as schematically illustrated in Fig. 14).[170] Such a novel hybrid configuration facilitates high mode overlap between the waveguide and the active material. As a result, the proposed devices show a low on-chip insertion loss (2.5 dB), a large EO bandwidth (>70 GHz), a high modulation efficiency (a voltage-length product of 2.2 V cm), and a high modulation rate (on-off keying modulation up to 100 Gbit/s and four-level pulse amplitude modulation up to 112 Gbit/s). The achieved excellent optical modulation characteristics therein are benefiting from efficiently utilizing the LN membrane, overcoming the voltage-bandwidth trade-off in the previous studies.[170] Following the similar fabrication and design strategy, a hybrid Si/LN Michelson interferometer modulator (MIM) with a reduced half-wave voltage-length product ($V_\pi \cdot L$) as low as 1.2 V cm has been demonstrated, representing the thus-far lowest $V_\pi \cdot L$ in

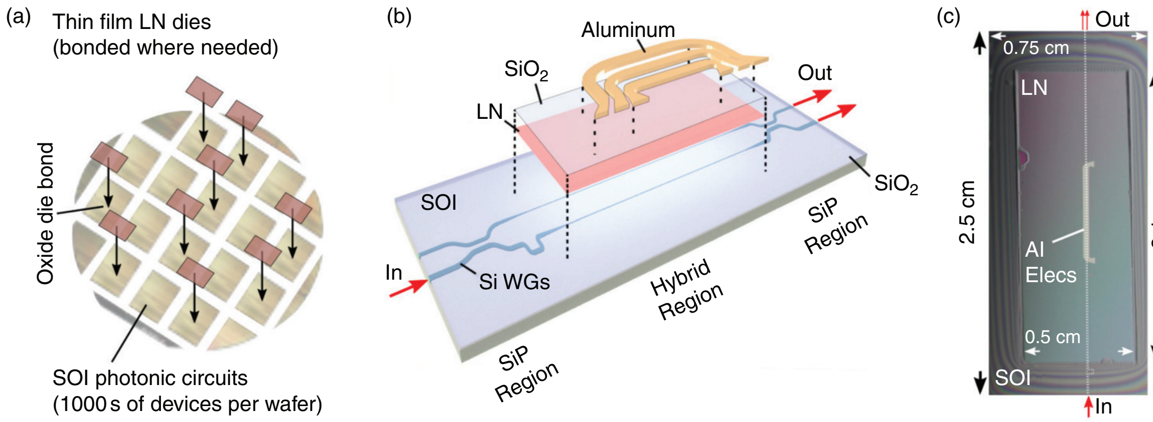

FIG. 13. (a) Thin film *x*-cut $LiNbO_3$ dies are bonded to segmented dies of a patterned and planarized silicon-on-insulator wafer, which contains fabricated silicon photonic waveguide circuits. (b) Exploded schematic of a heterogeneous Si/$LiNbO_3$ electro-optic modulator, where an unpatterned, un-etched LN thin film is bonded to a Mach–Zehnder interferometer fabricated on the silicon-on-insulator wafer. (c) Top view of a hybrid Si/$LiNbO_3$ electro-optic modulator.[168] Reproduced with permission from P. O. Weigel, J. Zhao, K. Fang, H. Al-Rubaye, D. Trotter, D. Hood, J. Mudrick, C. Dallo, A. T. Pomerene, A. L. Starbuck, C. T. DeRose, A. L. Lentine, G. Rebeiz, and S. Mookherjea, "Bonded thin film lithium niobate modulator on a silicon photonics platform exceeding 100 GHz 3-dB electrical modulation bandwidth," Opt. Express **26**(18), 23728–23739 (2018), Copyright 2018 Optical Society of America.

thin-film LN modulators.[171] Besides active photonic devices based on Si/LN platform, there has also been effort toward Si_3N_4/LN-based hybrid photonics, in which thin-film LN with an etched terrace structure is bonded to Si_3N_4 waveguides. Such a hybrid configuration significantly improves modal transition from the Si_3N_4 waveguide to the hybrid Si_3N_4/LN waveguide.[172] More recently, one of the most CMOS-compatible EO modulators has been demonstrated based on hybrid Si/LN photonics (via bonding of thin-film *x*-cut LN onto Si chip at low temperature) with buried Si-rich SiN_x waveguides, in which all the fabrication, including the radio-frequency (RF) and optical structures, are operated completely in CMOS facilities.[173]

Apart from hybrid waveguides and WGRs, high-quality on-chip 1D PhC nanobeam cavities and 2D PhC slabs have also been successfully fabricated based on the LN on Si platform.[50,51,59,134] The demonstrated high optical *Q*s (as high as 10^6 has been achieved), together with tight mode confinement, result in many highly enhanced, intriguing photonic phenomena, such as a significant cavity resonance tuning induced by the photorefractive effect (purely a classical phenomenon with a resonance tuning rate of 88 MHz/photon or, equivalently, 0.67 GHz/aJ, three orders of magnitude greater than other LN resonators),[51,134] a strong saturation and quenching of photorefraction (exhibiting a peculiar anisotropic behavior),[50,134] a strong coupling between the optical cavity modes and the mechanical motion of the PhC structures (mechanical modes with frequencies up to 1.003 GHz with an $f \cdot Q$ product of 1.47×10^{12} Hz),[134] a unique thermo-optic nonlinear behavior (which is considered to be a result of the pyroelectricity of LN),[50] and high-speed LN on-chip PhC-based EOMs (with a modulation bandwidth of 17.5 GHz, a tuning efficiency

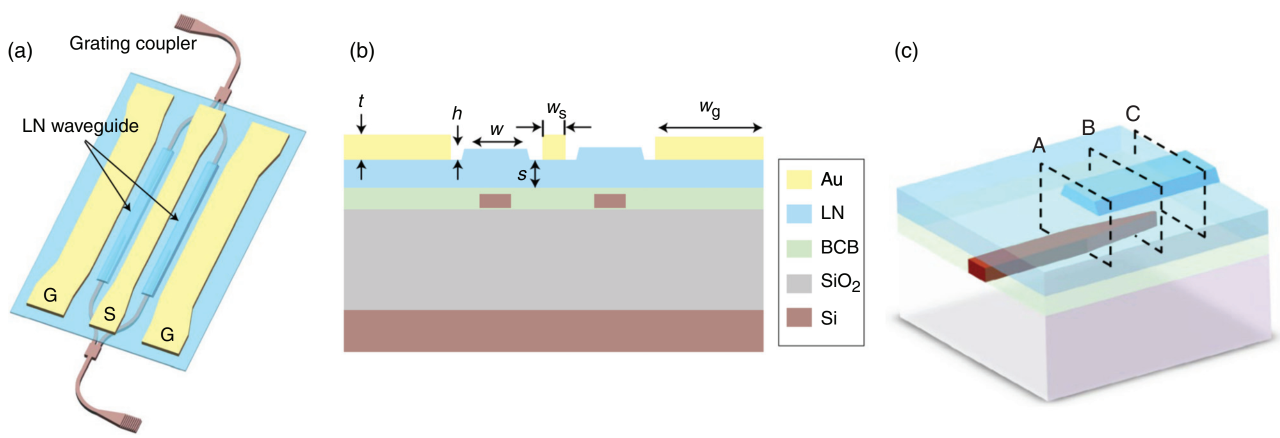

FIG. 14. (a) Schematic of the structure of the hybrid Si/$LiNbO_3$ Mach–Zehnder modulators demonstrated in the work reported by He *et al.* (b) Schematic of the cross section of the hybrid waveguide. (c) Schematic of the vertical adiabatic couplers.[170] Reproduced with permission from M. He, M. Xu, Y. Ren, J. Jian, Z. Ruan, Y. Xu, S. Gao, S. Sun, X. Wen, L. Zhou, L. Liu, C. Guo, H. Chen, S. Yu, L. Liu, and X. Cai, "High-performance hybrid silicon and lithium niobate Mach–Zehnder modulators for 100 Gbit s^{-1} and beyond," Nat. Photonics **13**(5), 359–364 (2019), Copyright 2019 The Author(s), under exclusive license to Springer Nature Limited.

up to 1.98 GHz V^{-1}, and an EO modal volume as small as 0.58 μm^3).[59] These promising integrated devices show great potential of hybrid LN PhC structures in a wide range of applications including optoelectronics, optomechanics, and electromechanics.

2. Rib-loaded LN on-chip photonic devices

In order to enhance the interaction strength (overlapping) of the waveguide mode and the active material, a more commonly used strategy is to define waveguide structures directly in the active layer rather than using only the evanescent tail of the waveguide mode. Taken in this sense, a platform hosting high-quality LN photonic devices and circuits on CMOS-processing-compatible substrates is more desirable, as first demonstrated by Fathpour's group at the University of Central Florida in 2013 [as schematically illustrated in Fig. 15(a)].[95] Based on wafer bonding and selective oxidation of refractory metals (tantalum) therein, LN on-chip microring optical resonators with a Q-factor of 7.2×10^4 and Mach–Zehnder optical modulators with a $V_\pi \cdot L$ of 4 V·cm have been demonstrated on silicon substrates. Particularly, Ta_2O_5 is used as a rib-loaded layer in combination with the LN thin film to form composite ridge waveguides and resonators. Alternatively, in the following studies, heterogeneous EO microring resonators (with a $Q = 1.85 \times 10^5$ and a 1.78 pm/V resonance tunability), thin-film MZMs (with a $V_\pi \cdot L$ of 3 V·cm and a 3-dB electrical bandwidth of 33 GHz), and PPLN waveguide nonlinear frequency converters (with a conversion efficiency of 160%W^{-1}·cm^{-2}) have been successively demonstrated by rib-loading and subsequent structuring of chalcogenide glass ($Ge_{23}Sb_7S_{70}$) or silicon nitride (SiN_x) layers on LN thin films.[144,174–179] This approach is therefore becoming a key technology for heterogeneous integration of LN photonic devices, offering new properties that can advantageously complement Si photonics. Actually, rib-loading thin-film LN with an easy-to-process material (maybe unconventional materials that are not CMOS-process compatible, such as $Ge_{23}Sb_7S_{70}$ and Ta_2O_5) is one of the most straightforward routes to achieving highly confined waveguides, as long as the refractive index of the deposited material matches that of the LN such that a large portion of optical energy is still confined within the waveguide core region. Most recently, in particular, a sub-1-volt V_π

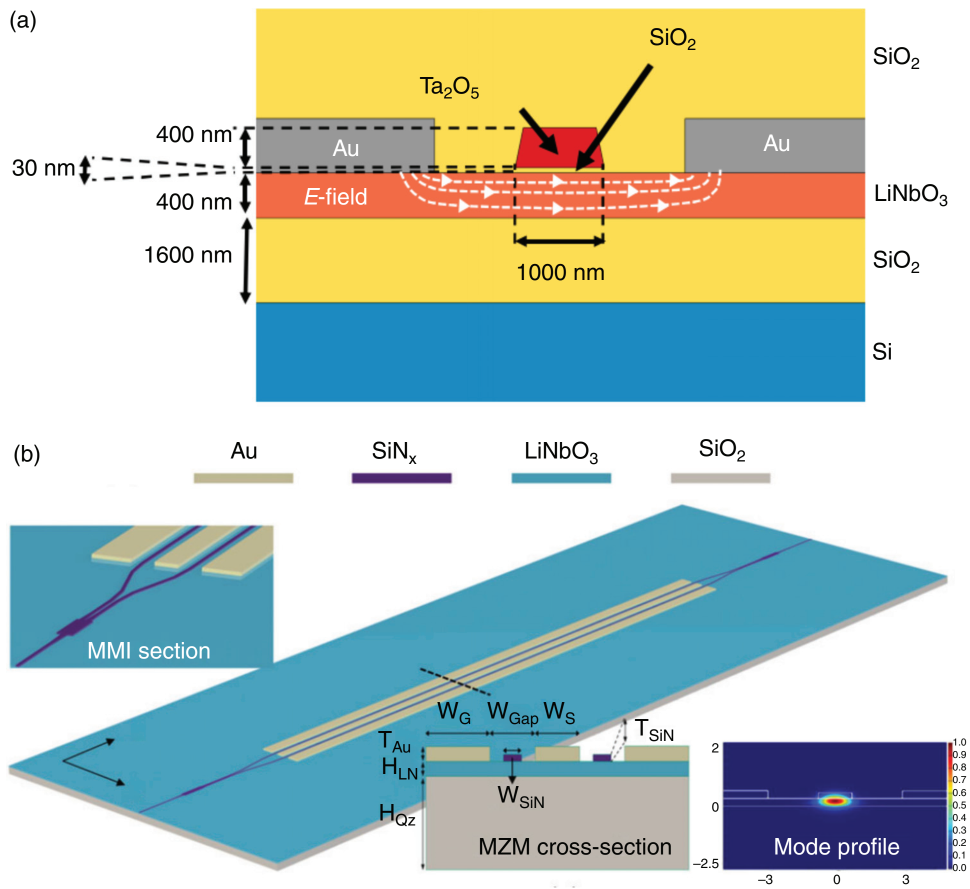

FIG. 15. (a) Schematic of the rib-loaded $LiNbO_3$ waveguide structure at one arm of the modulator reported in Rabiei *et al.*[95] Reproduced with permission from P. Rabiei, J. Ma, S. Khan, J. Chiles, and S. Fathpour, "Heterogeneous lithium niobate photonics on silicon substrates," Opt. Express **21**(21), 25573–25581 (2013), Copyright 2013 Optical Society of America. (b) Schematic of the thin-film modulator layout with hybrid $LiNbO_3$/SiN_x Mach–Zehnder modulators and electrodes fabricated in the work presented by Ahmed *et al.*[180] Reproduced with permission from A. N. R. Ahmed, S. Nelan, S. Shi, P. Yao, A. Mercante, and D. W. Prather, "Subvolt electro-optical modulator on thin-film lithium niobate and silicon nitride hybrid platform," Opt. Lett. **45**(5), 1112–1115 (2020), Copyright 2020 Optical Society of America.

($V_\pi = 0.875$ V, which is so far the lowest V_π shown for both standalone and hybrid thin-film LN-based EO modulators) MZM with a push-pull scheme has been demonstrated based on rib-loading of SiN_x on LNOI platform, as schematically illustrated in Fig. 15(b).[180] The subvolt V_π value therein is achieved by increasing the waveguide length (with a 2.4-cm-long interaction region) and reducing the electrode gap (5.8 μm) while maintaining the high electrical and optical field overlap at the LN region.[180] Apart from fabrication of on-chip active photonic devices, passive photonic structures such as heterogeneous waveguides and grating couplers can be as well defined applying the same strategy. For example, Si grating couplers (with a coupling efficiency of −18 dB at telecommunication band) on a Si-rib-loaded LNOI waveguide has been demonstrated.[181,182]

3. Heterogeneous integration of plasmonic structures

Besides heterogeneous integration of photonic structures made of dielectric/semiconductor materials, metal nanostructures, with their unique surface plasmon polaritons (SPPs) excitation at the metal/dielectric interface, are able to transmit as well as enhance both optical and electrical local fields, facilitating plasmonics in becoming a versatile platform for exceptionally compact optoelectronic applications.[183,184] Therefore, a proper integration of the unique plasmonic features on high-efficiency EO material platforms, especially LN (or LNOI), is expected to be a compact solution toward improved EO performance. Most recently, novel plasmonic EO directional coupler switches consisting of two closely spaced nm-thin gold nanostripes on LN substrates have been demonstrated [an SEM view is shown in Fig. 16(a)], enabling ultra-compact switching and modulation functionalities [as shown in Figs. 16(b)–16(d)].[185] The drive voltage-length product of 0.3 V·cm at telecom wavelengths has been realized, which is to-date the lowest value for any type of LN-based EOM. And such a strategy can be well adopted and imitated on LNOI-based platform with higher integration level.

4. Integrated quantum emitters

In systems utilizing optical and photonic components, quantum photonic emitters (quantum light sources that provide single photons) are the fundamental building blocks for a variety of quantum technologies, including quantum network, connecting quantum computing, quantum communication, and quantum metrology.[79,186,187] Furthermore, in the domain of integrated photonics, integrated on-chip quantum emitters are very promising for complex, scalable quantum photonic circuits.[79] In particular, LNOI technology provides the intrinsic strong EO properties of LN as well as the good mode confinement, which are both highly desired as an active platform for quantum photonics. Since LN lacks efficient single-photon emitters, heterogeneous integration and effective coupling of single-photon emitters, e.g., quantum dots, with LN photonic chips are required. Recently, quantum emitters of InAs quantum dots coupled with a nanophotonic LNOI waveguide has been demonstrated.[113] The LN waveguide and InP nanobeam cavity are fabricated by "EBL + dry-etching" on separate substrates. The so-called pick-and-place technique is then employed for transferring the nanobeam to the waveguide. The fabricated device is composed of a partially etched LNOI on-chip ridge waveguide (or a y-branch beam splitter), an InP nanobeam cavity with embedded InAs quantum dots on top of the waveguide, and an adiabatic taper for smoothly transferring the single photon to the waveguide [see Fig. 17(a)].[113] Despite being complicated, such a design is capable of high-efficiency coupling (evanescent field coupling with efficiency exceeding 34%) of single photons to straight waveguides and y-branch beam splitters, as verified by the measured photoluminescence signals shown in Figs. 17(b)–17(d),[113] suggesting a great potential of LNOI technology in implementing fast, reconfigurable quantum photonic circuits.

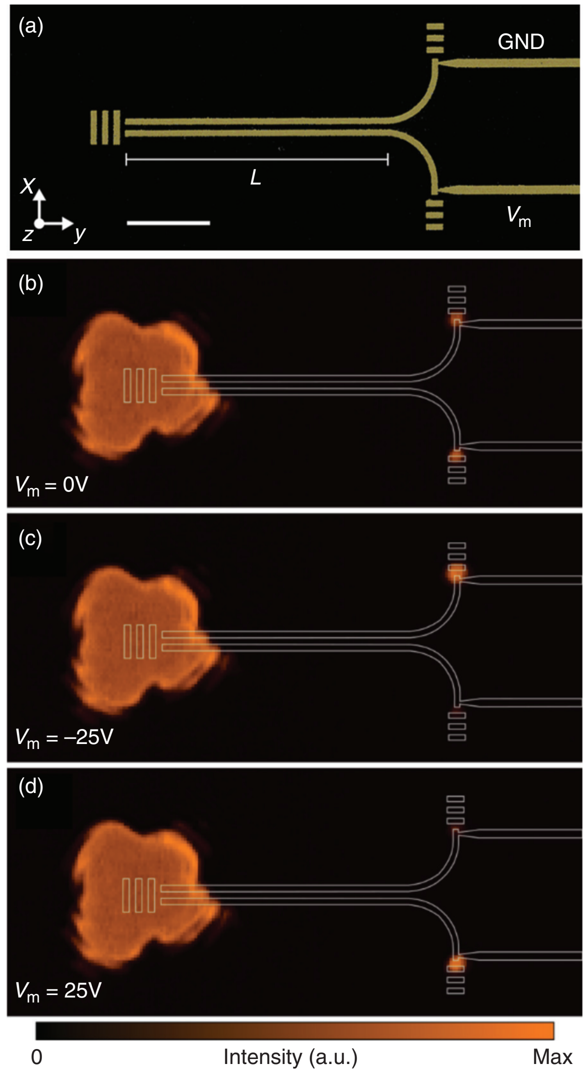

FIG. 16. (a) False-colored scanning electron microscopic image of the $LiNbO_3$/plasmonic directional coupler switch reported by Thomaschewski *et al.*[185] Experimental optical far-field images captured by an infrared camera at different modulation voltages: (b–d) $V_m = 0$, −25, and 25 V. Reproduced with permission from M. Thomaschewski, V. A. Zenin, C. Wolff, and S. I. Bozhevolnyi, "Plasmonic monolithic lithium niobate directional coupler switches," Nat. Commun. **11**(1), 748 (2020), licensed under a Creative Commons Attribution 4.0 International License, Copyright 2020 The Author(s).

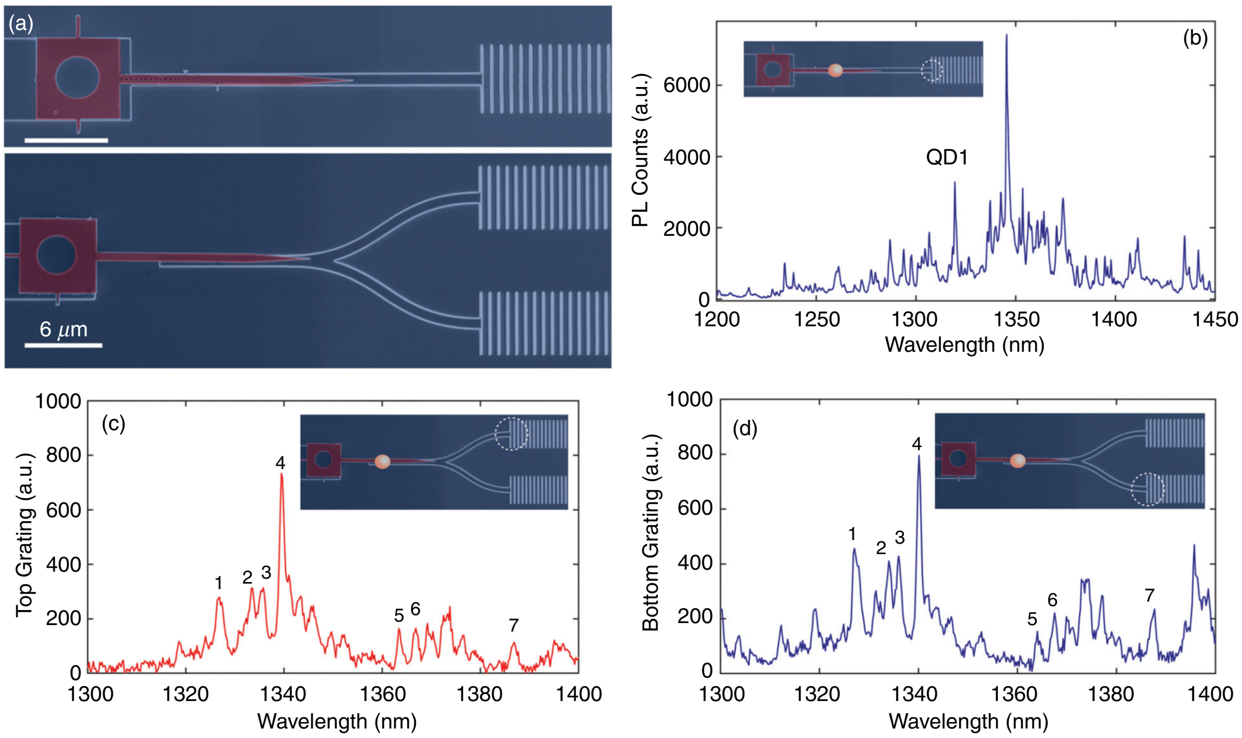

FIG. 17. (a) False color scanning electron microscopic images of the integrated InP nanobeam with a straight $LiNbO_3$ on-chip waveguide and an on-chip beam splitter reported by Aghaeimeibodi *et al.*[113] (b) Photoluminescence spectrum of the $LiNbO_3$ on-chip waveguide coupled quantum dots. The inset indicates the excitation and collection scheme, in which the orange dot and the white dashed circle represent the excitation and collection spots, respectively. Photoluminescence spectra collected through the (a) top and (b) bottom gratings from the $LiNbO_3$ on-chip *y*-branch beam splitters. Insets indicate the excitation and collection scheme. Reproduced from S. Aghaeimeibodi, B. Desiatov, J.-H. Kim, C.-M. Lee, M. A. Buyukkaya, A. Karasahin, C. J. K. Richardson, R. P. Leavitt, M. Lončar, and E. Waks, "Integration of quantum dots with lithium niobate photonics," Appl. Phys. Lett. **113**(22), 221102 (2018), with the permission of AIP Publishing.

5. Integrated optical metasurface structures

In LNOI on-chip optical devices, a nonlinear phase matching condition is, in general, achieved using either birefringence or periodic inversion of ferroelectric domains, both of which require additional dispersion engineering and are in general narrowband. Aiming at this limitation, optical metasurface structures, which generally consist of periodically distributed nanoantennas with subwavelength separation, are introduced into LNOI integrated photonics to circumvent the phase-matching requirement.[111] The basic strategy is to pattern a gradient metasurface structure on the top surface of an on-chip waveguide, as schematically shown in Fig. 18(a),[112] such that the gradient metasurface is able to control guided waves by strong, consecutive scattering events at the antenna array via evanescent field coupling. As a result, with a proper design of the antenna array and the waveguide structure, the guided light propagation as well as the waveguide modes can be controlled at will, allowing for the realization of a non-perfect phase-matching condition. A significant advantage of this strategy for applications in nonlinear optics is that it supports optical components along both TE and TM polarizations. Figure 18(b) shows the working principle of the metasurface-based nonlinear integrated photonic devices.[112] In the metasurface-patterned waveguide region, the optical power first couples from the fundamental mode $TE_{00}(\omega)$ at the pump frequency to the fundamental mode $TE_{00}(2\omega)$ at the SH frequency, and then to the higher-order waveguide modes $TE_{mn}(2\omega)$ and $TM_{mn}(2\omega)$ at the SH frequency with the assistance of the gradient metasurface. The unidirectional wavevector provided by the gradient metasurface ensures that coupling of optical power back from the $TE_{mn}(2\omega)$ and $TM_{mn}(2\omega)$ modes to the $TE_{00}(2\omega)$ mode or the $TE_{00}(\omega)$ mode is highly inefficient. Such a unidirectional optical power transfer results in an effective accumulation of SHG power as a function of the propagation distance, namely, the so-called phase-matching-free nonlinear generation, by which the conversion efficiency is sensitive to the variation of neither pump frequency nor device geometry. Recently, on-chip devices operating under such a new scheme are demonstrated by patterning phased arrays of *a*-Si nano-rod antennas on nanophotonic LNOI waveguides (the devices were fabricated by employing a combination of EBL, RIE, and PECVD techniques), exhibiting an efficient SHG with a normalized conversion efficiency of 1660% $W^{-1}cm^{-2}$ over a broad range of pump wavelengths (over 1.5–1.6 μm).[112] Subsequently, based on a grating metasurface patterned on the top surface of a well-designed LNOI on-chip waveguide (fabricated by FIB), SHG as well as in-plane manipulation of the generated SH wave is demonstrated. The latter is realized by using a 90°-turn reflection Bragg grating.[188] Furthermore, in order to manipulate the wavefront of the SH signals inside the waveguide, a 2D hologram is encoded in the grating metasurfaces. Thus, strong

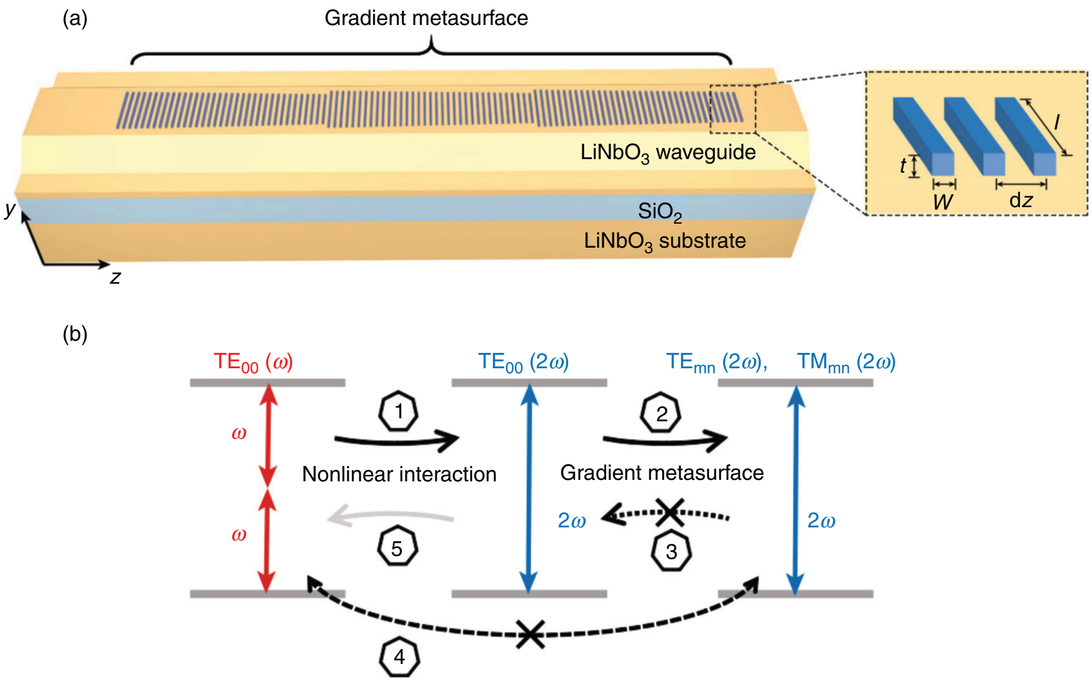

FIG. 18. (a) Schematic of the $LiNbO_3$ on-chip ridge waveguide integrated with a well-designed gradient metasurface for achieving phase-matching-free second harmonic generation as demonstrated by Wang *et al.*[112] (b) Conceptual diagram of the metasurface-based phase-matching-free second harmonic generation. Reproduced with permission from C. Wang, Z. Li, M. H. Kim, X. Xiong, X. F. Ren, G. C. Guo, N. Yu, and M. Lončar, "Metasurface-assisted phase-matching-free second harmonic generation in lithium niobate waveguides," Nat. Commun. **8**(1), 2098 (2017), licensed under a Creative Commons Attribution 4.0 International License, Copyright 2017 The Author(s).

functionalities of nonlinear beam shaping, including dual focusing and Airy beam generation, are demonstrated based on the same device, suggesting the great potential of optical metasurfaces for integrated nonlinear nanophotonics.

6. Bound states in the continuum (BICs)

Bound states in the continuum (BICs) are a special but general wave phenomenon, lying in the continuous spectrum but exhibiting as spatially localized states.[189] In photonics, such a unique feature allows for an extremely strong optical confinement and thus a very high optical enhancement of the localized light field. Benefiting largely from the tremendous progress of advanced nanofabrication technologies, research of photonic BICs has experienced a rapid growth in terms of scope and practicability in recent years.[190–193] So far, a wide spectrum of BICs-based photonic applications such as nanostructured sensors, lasers, and filters have been demonstrated, rendering this concept a novel solution for diversifying and enhancing photonic functionalities for integrated optics.[194–196] Particularly, the desired low-loss light guidance in BICs-based PICs can be realized by means of patterning a low-refractive-index waveguide on the surface of a high-refractive-index substrate.[197,198] In such a configuration, the light field is confined within a region of the substrate below the waveguide, which is highly compatible with the rib-loading technique for fabrication of LNOI on-chip photonic devices, only with loading materials that possess a refractive index lower than, instead of equal to, that of LN. Therefore, harnessing BICs in PICs circumvents the stringent requirement of direct etching as well as introduces more functional optical materials into integrated optics, including but not limited to LN. Recently, a high-quality ($Q > 500{,}000$) photonic microcavity based on the BIC mechanism has been demonstrated on a LNOI platform via patterning a rib-loaded low-refractive-index racetrack-shaped polymer waveguide, as schematically illustrated in Fig. 19(a).[199] The on-chip microcavity is also monolithically integrated with an surface acoustic wave (SAW) interdigital transducer (IDT), efficiently supporting SAW excitation and propagation in the unetched LN thin film region. As a result, acousto-optic modulation (with modulation frequencies beyond 4 GHz) of cavity-enhanced photonic BIC mode has been demonstrated, facilitating highly efficient phonon-photon coupling on a chip as evidenced by the demonstrated electro-acousto-optically induced transparency and absorption, as shown in Fig. 19(b).[199] The demonstrated results of manipulating and controlling photonic BICs on a LNOI chip indicate extensive applications of such a scheme in microwave photonics and quantum information processing. Furthermore, most recently, BIC modes with different orders have been experimentally excited in well-designed waveguides/directional couplers (BICs can be obtained for the TM bound modes in each order just by engineering the waveguide width).[200] The fabricated multimode directional couplers (including TM_0-TM_1, TM_0-TM_2, and TM_0-TM_3) exhibit insertion losses <1.5 dB in the wavelength range of 1.51–1.58 μm, with the extinction ratio >13 dB. By further cascading the multimode directional couplers, a four-channel TM mode (de)multiplexer (which is composed of TM_0, TM_1, TM_2, and TM_3 mode channels) is constructed on the LNOI platform,

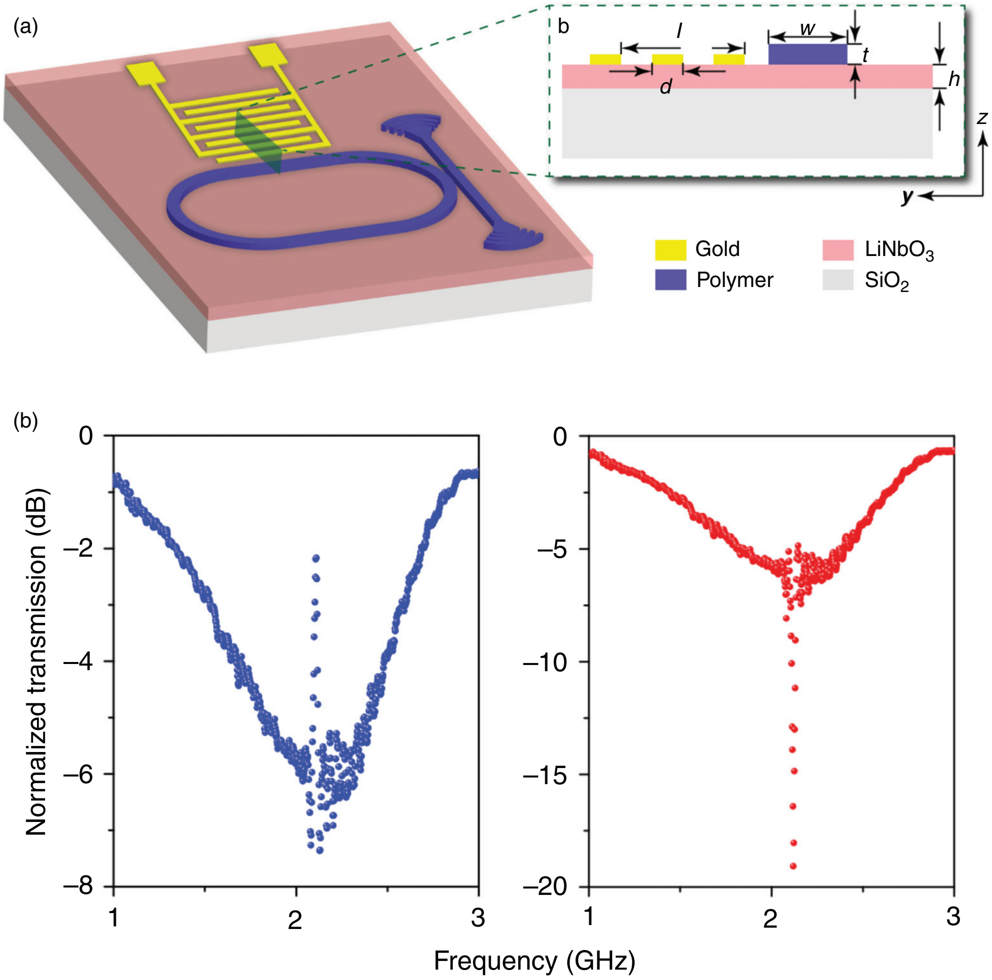

FIG. 19. (a) Schematic of the photonic microcavity based on the bound-states-in-the-continuum mechanism demonstrated by Yu *et al.*,[199] which is constructed from a low-refractive-index racetrack-shaped polymer waveguide on a high-refractive-index $LiNbO_3$ thin film. The surface acoustic wave interdigital transducer made of Au electrodes is placed near a straight section of the racetrack microcavity. (b) The coherent coupling between microwave and optical photons exhibited by electro-acousto-optically induced transparency and absorption measured by Homodyne measurement setup. Reproduced with permission from Z. Yu and X. Sun, "Acousto-optic modulation of photonic bound state in the continuum," Light: Sci. Appl. **9**(1), 1–9 (2020), licensed under a Creative Commons Attribution 4.0 International License, Copyright 2020 The Author(s).

demonstrating an insertion loss <4.0 dB and a crosstalk <−9.5 dB in the telecom wavelength range. By exploiting the four-channel TM mode (de)multiplexer, high-dimensional optical communication with a data transmission at 40 Gbps/channel has been demonstrated, suggesting huge potential of harnessing BIC modes in hybrid integrated photonic platforms, such as LNOI, for constructing high-capacity, high-speed on-chip optical communication.[200] By further integrating with EOMs, such a mode (de)multiplexer is capable of providing both EO modulation and mode (de)multiplexing on a single chip.[200]

In summary, great effort has been made toward integrating a higher number of new optical materials and nanostructures onto LNOI photonic chips for high performance devices as well as completely new functionalities. While the whole field might have started with the integration of electrodes for the realization of LNOI on-chip EO tuning and modulation devices, the diversity of functionalities and integrated materials envisaged has widely expanded and is now on the verge of overcoming the long-lasting difficulty of integration of on-chip light sources. New functions/applications as well as elegant solutions at both design and fabrication have arisen along the way as hybrid LNOI photonics technologies continue to develop, creating new opportunities for advanced hybrid devices and circuits.

D. Optical coupling strategies

A major challenge ever-present for practical applications of LNOI thin films in PICs is high-efficiency optical coupling between the standard optical fibers and micro-/nano-scale on-chip photonic devices. An efficient optical coupler/interface is of significant importance for many photonic applications, such as EO modulation and nonlinear optical frequency conversion, whose overall performance is largely determined by the total in-coupled optical power. In the scenario of end-face coupling, special designs of the coupling fiber end-facets (such as lensed optical fibers[101] and tapered optical fibers[123]) are usually required due to the large mode mismatch between standard optical fibers and LNOI on-chip photonic devices. However, the fiber-to-chip loss is still relatively high (generally 5–10 dB/facet).[18,23,26] Most recently, by meticulously studying the geometric parameters of the LNOI on-chip waveguide and fiber taper, fiber-to-chip coupling loss reduced to 1.32 dB (for TE mode, 1.88 dB for TM mode) is demonstrated, the SEM view of the fiber taper tip is shown in Fig. 20(a).[201] This value also represents the lowest fiber-to-chip coupling loss for LNOI photonics so far. Such an efficient technique is based on adiabatically tapering a standard single-mode fiber (the cross-sectional profile of the fiber is precisely controlled by a fiber-pulling system) until the mode of the fiber sufficiently matches the mode of, in particular, a designed LNOI on-chip waveguide. The development of this technique is expected to be quite beneficial for a broad range of LNOI-based PIC applications.

In contrast to end-face coupling that only occurs at the fiber and device end-facets, grating couplers enables optical coupling anywhere on the thin film surface with good optical alignment tolerance, although generally with a relatively low coupling efficiency. For example, the coupling losses of dry-etched LNOI on-chip grating couplers (integrated with on-chip waveguides) in the pioneering studies were determined to be >10 dB.[202,203] Subsequently, it has been found that by adding a bottom reflector layer (such as a metal bottom reflector) in the LNOI, the power leakage can be minimized and thus the coupling efficiency can be enhanced, reducing the coupling loss from 9.1 dB to 6.9 dB in experiments.[203] In order to further reduce the fiber-to-chip coupling loss of grating couplers integrated on LNOI system, a lot of attempts have been made on parameter optimization and configuration design. For example, by means of systematically studying the contribution (the loss is mainly induced by modal mismatch and reflections) to the coupling loss of grating couplers and adopting a

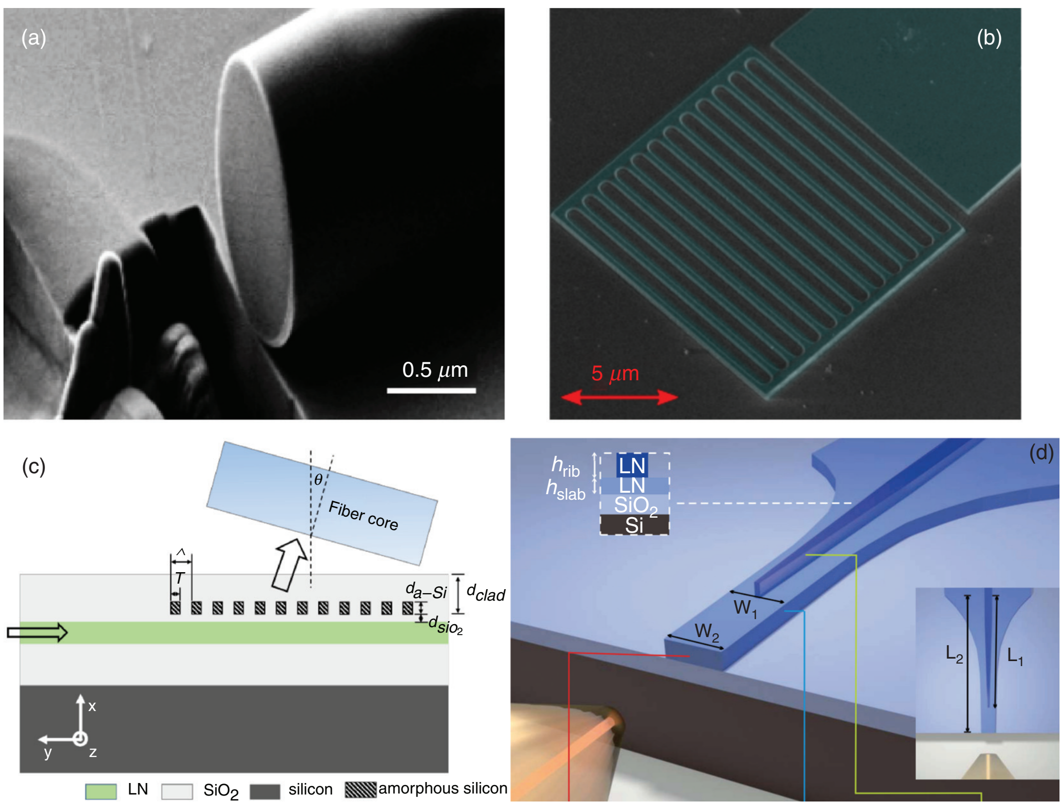

FIG. 20. (a) Scanning electron microscopic view of the tip of the fiber taper after focused ion beam process as reported by Yao *et al.*,[201] giving a fiber-to-chip coupling loss as low as 1.32 dB. Reproduced with permission from N. Yao, J. Zhou, R. Gao, J. Lin, M. Wang, Y. Cheng, W. Fang, and L. Tong, "Efficient light coupling between an ultra-low loss lithium niobate waveguide and an adiabatically tapered single mode optical fiber," Opt. Express **28**(8), 12416–12423 (2020), Copyright 2020 Optical Society of America. (b) Scanning electron microscopic view of the $LiNbO_3$ on-chip grating coupler designed and fabricated by Krasnokutska *et al.*,[205] featuring a coupling efficiency of 44.6%/grating. Reproduced with permission from I. Krasnokutska, R. J. Chapman, J. J. Tambasco, and A. Peruzzo, "High coupling efficiency grating couplers on lithium niobate on insulator," Opt. Express **27**(13), 17681–17685 (2019), Copyright 2019 Optical Society of America. (c) Schematic of the *a*-Si grating on $LiNbO_3$ thin film, providing a coupling loss as low as 3.06 dB as reported by Jian *et al.*[182] Reproduced with permission from J. Jian, P. Xu, H. Chen, M. He, Z. Wu, L. Zhou, L. Liu, C. Yang, and S. Yu, "High-efficiency hybrid amorphous silicon grating couplers for sub-micron-sized lithium niobate waveguides," Opt. Express **26**(23), 29651–29658 (2018), Copyright 2018 Optical Society of America. (d) Schematic of the bilayer tapered mode size converter, exhibiting a coupling loss as low as 1.7 dB/facet as demonstrated by He *et al.*[206] Reproduced with permission from L. He, M. Zhang, A. Shams-Ansari, R. Zhu, C. Wang, and L. Marko, "Low-loss fiber-to-chip interface for lithium niobate photonic integrated circuits," Opt. Lett. **44**(9), 2314–2317 (2019), Copyright 2019 Optical Society of America.

chirped grating coupler in the LNOI system, a coupling loss of 3.6 dB/grating is demonstrated, which exhibits a 1.9-dB improvement with respect to a uniform grating coupler.[204] Similarly, a measured coupling efficiency of 44.6%/grating (for TE polarization, corresponding to 3.5 dB/grating) is achieved based on detailed investigation of grating coupler design parameters through simulation and experiment [the SEM view of which is shown in Fig. 20(b)].[205] Alternatively, based on a hybrid platform, an α-Si grating coupler on LNOI is able to exhibit a coupling loss as low as 3.06 dB with a 1-dB bandwidth of 55 nm,[182] benefiting from the high refractive index and the shaper etch profile of α-Si in contrast to that of LN. Figure 20(c) schematically illustrates the hybrid design.

Most recently, a dry-etched, on-chip, bilayer, inversely tapered, mode size converter [consists of a two-layer taper with gradually varied heights, as schematically illustrated in Fig. 20(d)] was designed and fabricated based on LNOI platform, exhibiting an extremely low coupling loss of 1.7 dB/facet.[206] At a certain point in the gradually varied waveguide cross section, good mode overlap and index match between modes of input fibers and on-chip waveguides can be achieved. The combination of such a compact coupler design and a high coupling efficiency is highly beneficial for optical packaging of LNOI devices and practical applications of LNOI platform.

V. CONCLUSIONS AND PERSPECTIVES

The past few years have witnessed a number of technological breakthroughs in the fabrication, structuring, and engineering techniques toward the realization of advanced PICs, facilitating the booming development of LNOI technology. In this article, the recent technological developments of LNOI platform for photonic integration and performance optimization of on-chip devices have been reviewed. It has been shown that many of the key technologies with a view to achieving highly integrated and functional LNOI PICs have been established. The most important step toward practical applications is the production of high-quality, large-scale LNOI wafers, which have been commercially available as the ion cut and wafer bonding technologies mature. Most recently, monolithic LN PICs (based on 4- and 6-inch wafers) with low optical propagation loss (0.27 dB/cm), a good uniformity, and a high throughput have been demonstrated, representing a first step toward large-scale, complex, and low-loss EO and nonlinear PICs with high yield.[207] In parallel, it is also stimulating emerging large-scale PIC applications such as quantum photonics and photonic neural networks.[79,208] Even so, there is still tremendous interest to further reduce the loss to the state-of-the-art at the single device level. So far, by employing an optimized ICP-RIE process and a subsequent sidewall fine polishing or the fs-laser-assisted CMP process, LNOI on-chip waveguides with an optical propagation loss of as low as 2.7 dB/m have been demonstrated in both cases.[99,102,104,106–108] Despite the fact that moving in the direction toward absorption-limited (or to say, achieving single-mode waveguide with a propagation loss of as low as 1 dB/m level) LN PICs is full of challenges, further attempts can be made in how to combine properly different surface structuring and subsequent polishing techniques while maximizing their respective advantages based on the LNOI platform. In addition, systematic research on the fabrication parameter optimization of ion-cut LNOI technology with a view to enlarging the wafer size (to the 8-inch level or even larger) while maintaining the high quality of the optical, thermal, mechanical, and crystalline properties of the LN thin film are highly desired. Besides, the photonic functionalities and performance of LN devices in high-power applications are limited by LN's low-damage threshold, especially for ultrahigh-*Q* microcavities.[3,209] To overcome this shortcoming, an ideal alternative is $LiTaO_3$ (LT), which possesses optical properties very similar to that of LN but a higher optical damage threshold and a good UV transparency.[210] Recently, a high-quality ion-cut LT-on-insulator (LTOI) film has been prepared, demonstrating an on-chip microdisk with a *Q*-factor of 2.67×10^5 fabricated by FIB milling and a high optical damage threshold (on the order of 10^5 W/cm^2).[211] This work may pave the way for LTOI to find applications in future PICs.

An often-concerned point regarding LN is its photorefractive effect, which may result in device performance degradation or even severe crystal damage. However, there is experimental evidence indicating that the LNOI photonic platform shows much better resistance to photorefractive damage in contrast to their bulk counterparts,[46,55,134] which can be attributed to the fast relaxation time of photorefractive effect in the case of thin-film LN.[212] In addition, the refractive index contrast of LNOI on-chip waveguides is much larger than that can be induced by the photorefractive effect, enabling unaffected optical modes.[16] Additionally, refractive index change can be also induced by the thermal-optic effect. However, for LN, the directions and time scales of the induced refractive index change in the case of thermal and photorefractive effect are very different, leading to competition between these two effects and resulting in lots of interesting optical phenomena that are difficult to be stabilized at certain optical power levels.[212] It is worth mentioning that the photorefractive damage effect, in bulk as well as thin-film LN, can be significantly suppressed by doping of, for example, MgO and ZrO_2.[213,214] This strategy is highly compatible with the standard ion-cut LNOI technology and thus can be easily adopted in thin film LN.

On a different front, the continuously optimized surface structuring, engineering, and post-processing techniques are highly beneficial and promising for various LNOI on-chip structures and devices with excellent optical performance. These have enabled implementation of multifunctional LN-based systems on a monolithic chip, allowing for not only compact geometries but more importantly orders of magnitude performance improvements in, for example, integrated nonlinear frequency converters and optical frequency combs.[16,23–25,81]

An efficient LNOI on-chip optical coupler/interface is of significant importance for many photonic applications. Although a number of achievements and breakthroughs have been achieved,[182,201,204–206] further investigation and development of efficient and novel optical couplers/interfaces is still necessary. A potential way out may be imitating the successful technological advances that have been made on other CMOS-compatible material platforms such as SOI and SiN.[215,216] It is worth mentioning that free-space coupling without any optical couplers or interfaces is also possible in the case of deformed LNOI on-chip microcavities.[217] Apart from optical couplers/interfaces, polarization elements play important roles in integrated photonics as well. Thus far, an electrical controlled on-chip Solc-type device based on PPLNOI ridge waveguides (fabricated by diamond blade dicing) for TE↔TM mode conversion for polarization control and wavelength filtering with a low drive voltage and a wide operational wavelength has been experimentally demonstrated.[117] There are also a number of theoretical designs and proposals of LNOI on-chip polarization elements demonstrating broadband polarization beam splitters and polarizers, showing great potential of LNOI-based sub-wavelength devices for precise polarization management.[218–220]

A further important advance has been the recent availability of thin film materials that can be easily integrated on heterogeneous substrates. Advances in the deposition/growth of high-quality materials onto LNOI platform by Chemical vapor Deposition (CVD) or epitaxy have increased the number of functionalities that can be integrated onto LN. In addition, the tremendous development of diverse 2D layered materials in recent years has opened up many new possibilities and opportunities for hybrid integrated photonics, as these multifunctional nanomaterials can be flexibly integrated onto planar photonic platforms. In a hybrid integrated system with an ultrahigh integration level, 2D layered materials are expected to not only improve the original photonic performance of on-chip devices but also bring fundamentally new photonic functionalities and properties, such as biosensing, photovoltaics, photodetection, thermoelectrics, and so on into photonic chips.[221–225] A similar photonic integration strategy of 2D layered materials onto LNOI platform applies as well to 0D and 1D nanomaterials, such as quantum dots, nanowires, carbon nanotubes, and so on.[226–229]

Besides expanding the device functionality by adding new materials, hybrid LNOI photonics also aims to take advantage of the high optical confinement to harness the light properties at the micro- and nano-scale and thus enhance the light-matter interaction. Bound states in the continuum (as briefly discussed in Sec. IV),[189,198] topological photonics,[230–233] and non-reciprocal photonics[234,235] are just a few nanophotonic examples of the new concepts that help to better confine or guide light in chip-based platforms. In addition, 3D structure fabrication in LNOI using, e.g., femtosecond-laser micromachining or a stacked-integrated approach, such as the fabrication of double-layered LNOI on-chip microresonators,[70] holds great promise for device performance improvement and functionality enrichment by introducing a further dimension into the on-chip structures. This may be of great importance for practical application of LNOI platform in future 3D integrated microsystems.

So far, most demonstrations of LNOI on-chip photonic devices are operated in the telecommunications bands,[16,23–25] despite some remarkable achievements in the visible spectrum, such as fabrication of ultra-low loss waveguides and microresonators.[103,125] In particular, the expansion of LNOI photonics into the visible wavelengths could find a wide spectrum of applications, ranging from quantum optics and metrology to bio-sensing and biomedicine. And this can be one of the important research orientations in future LNOI photonics toward multi-functional PICs.

The study of LNOI-based quantum photonic platforms is still in its infancy, although LNOI on-chip waveguides integrated with quantum emitters and PPLNOI-based high-quality entangled photon pair generation have been demonstrated. In the near future, we foresee rapid progress in more complex and multicomponent quantum photonic circuits exploiting PPLNOI technology.[236,237] Furthermore, along with the efficient integration of quantum emitters with LNOI photonic circuits, a great deal of effort has to be made in controlling the quantum emitters, which is essential for quantum operation based on multiple indistinguishable single photons. Moreover, solid-state quantum nanophotonics has emerged as one of the most active fields in quantum engineering.[152] In particular, RE-doped materials, e.g., RE:LN, have attracted significant interest in recent years, enabling a new class of photonic devices that are highly coherent, efficient, and scalable, such as optical memories, photon sources, and optical-microwave transducers.[238–240] And these devices can be further miniaturized and integrated based on the RE:LNOI platform via a proper design, engineering, and fabrication. Generally speaking, implementation of scalable quantum photonic networks requires the integration of compatible photon sources and quantum memories.[241] As for the former, up until now, Er^{3+}:LNOI on-chip microresonator-based lasers have been demonstrated, showing great potential in PIC applications such as on-chip tunable laser sources and amplifiers.[159] While for the latter, RE:LN crystals have been used to implement solid-state quantum memories at different wavelengths.[9,151] To establish efficient quantum operation on a monolithic chip, the LNOI photonic circuits should combine many high-quality quantum devices/components with functionalities in routing, modulation, and detection of the generated photons, which can be enabled by the technical achievements in recent years. However, the integration of all required quantum devices/components on a monolithic photonic chip made by only one kind of optical material is far beyond the scope of any homogenous material platform, including LNOI. A sophisticated and modularized combination of different building blocks responsible for quantum light sources, filtering, optical delays, interference, active switching/modulation, detection, and control electronics based on heterogeneously integrated material systems may be the smart way out, indicating that monolithic and modular hybrid integration will be required.[242–248] Therefore, sophisticated hybrid integration, including not only different components but also different physical systems on one chip, can be envisioned.

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DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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